ES 544
Random Processes

with

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@

GIKI - FES

Probability

Continuing our discussion from the last lecture, we now look at…

- Quick Review of Common Distributions
- Law of Rare Events and the Poisson Distribution
- Central Limit Theorem and the Gaussian Distribution
  - Exponential Family
- A Recipe for Generating More Distributions

- Thus far, we have only looked at the distribution functions on an abstract level (general forms).

- What are some of the forms these functions often take (in practice)?

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- What are some of the forms these functions often take (in practice)?

    - This questions leads to our next topic



- Extracting Key Indicators (moments)
- Reviewing Common Distributions
- Estimating Parameters

Note: from here on, we will use $f(x)$ and $p(x)$ to represent both Probability Distribution (discrete X) and Probability Density (continuous X), with distinction made clear through context.

Discrete Case

- Uniform Distribution
- Binomial Distribution
- Geometric Distribution
- Negative Binomial Distribution
- Poisson Distribution

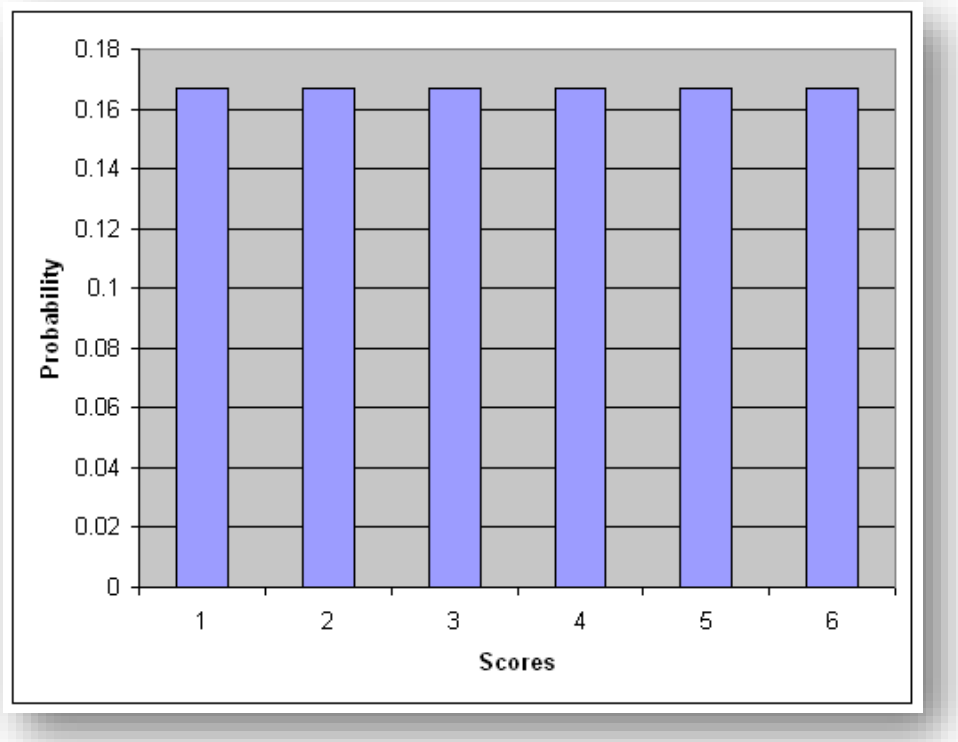
} Based on **Bernoulli Trials**

(There are, of course, many others)

Discrete Uniform Distribution

Discrete Uniform Distribution = when all possible outcomes have equal probabilities

- X = result of a rolling a die
- $x = 1, 2, 3, 4, 5, 6$ (possible values of X)
- All outcomes have equal probability
- $f(x) = \begin{cases} \frac{1}{6}, & x = 1,2,3,4,5,6 \\ 0\,, & otherwise \end{cases}$



Discrete Uniform Distribution

$$f(x) = \begin{cases} \frac{1}{n}, & x = x_1, x_2, x_3, \dots, x_n \\ 0, & otherwise \end{cases}$$

A random variable X has a discrete uniform distribution if

- It has a finite set of outcomes (say $x_1, x_2, x_3, \dots, x_n$), and
- Each outcome has exactly the same probability.
- The distribution of X with n equally probably outcomes may then be written as

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Interesting result and an alternate proposal for axioms of probability

- As per Kolmogorov's axioms of probability, a discrete random variable with countably infinite states cannot have a uniform distribution!
- HW: read up on the "De Finetti Lottery" and its implications for the philosophy of probability theory.

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Interesting result and an alternate proposal for axioms of probability

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If you set

$$P[X = k] = \alpha$$
$$k = 0, 1, 2, 3 \dots$$

Then

$$\sum_{k=0}^{\infty} \alpha = \begin{cases} \infty, & \alpha > 0 \\ 0, & \alpha = 0 \end{cases}$$

Both of these violate Kolmogorov's axioms

Discrete Uniform Distribution

Mean and Variance of a uniformly distributed random variable X

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Mean and Variance of a uniformly distributed random variable X

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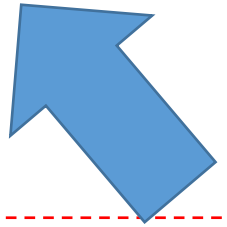
$$E[X] = \mu_X = \sum_x x f(x) = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$V[X] = \sum_x (x - \mu_X)^2 f(x) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu_X)^2$$

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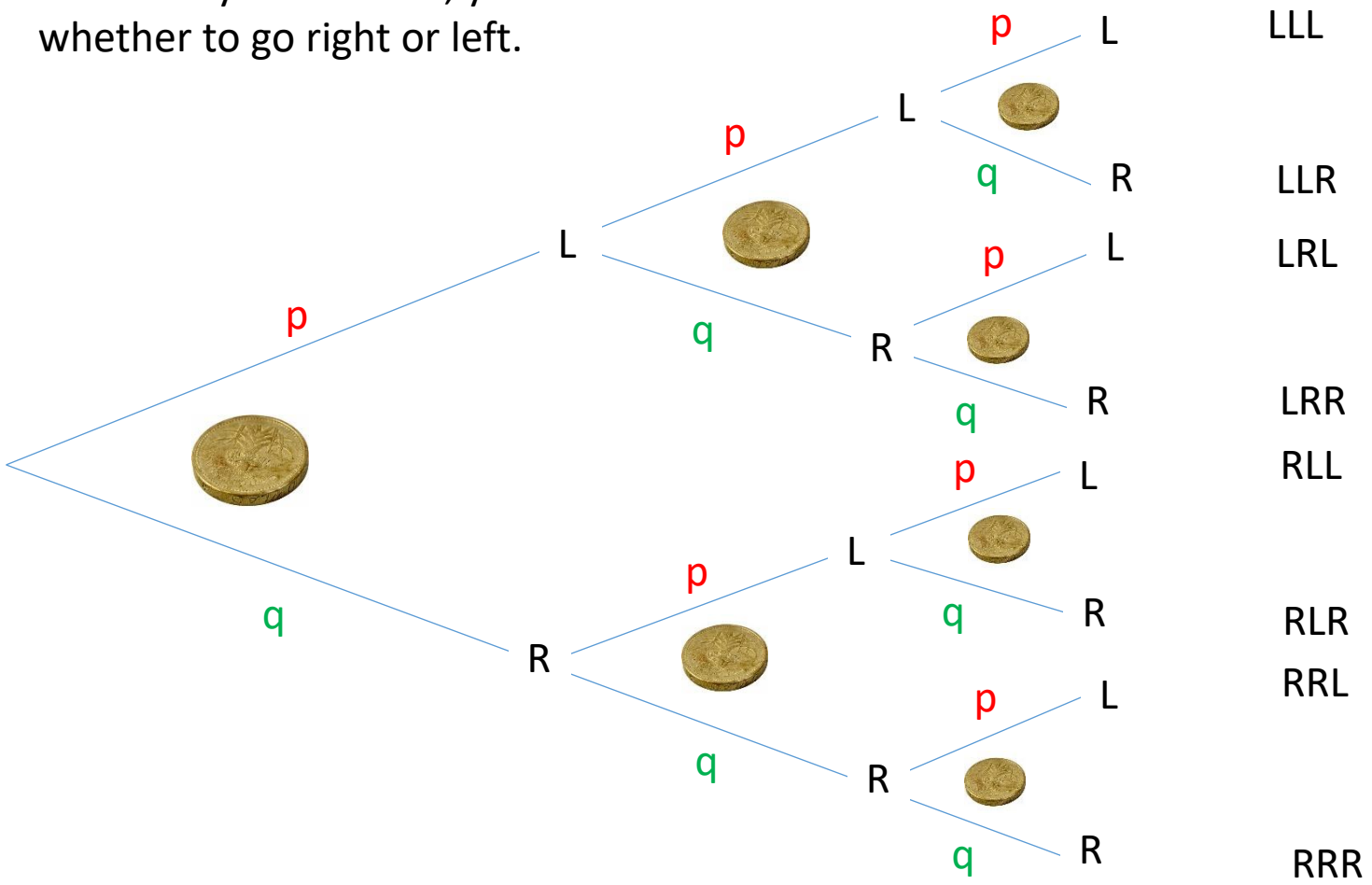
- Suppose you have plenty of free time (and petrol) and decide to go on a special drive.
- Each fork you arrive at, you toss a fair coin to decide whether to go right or left.





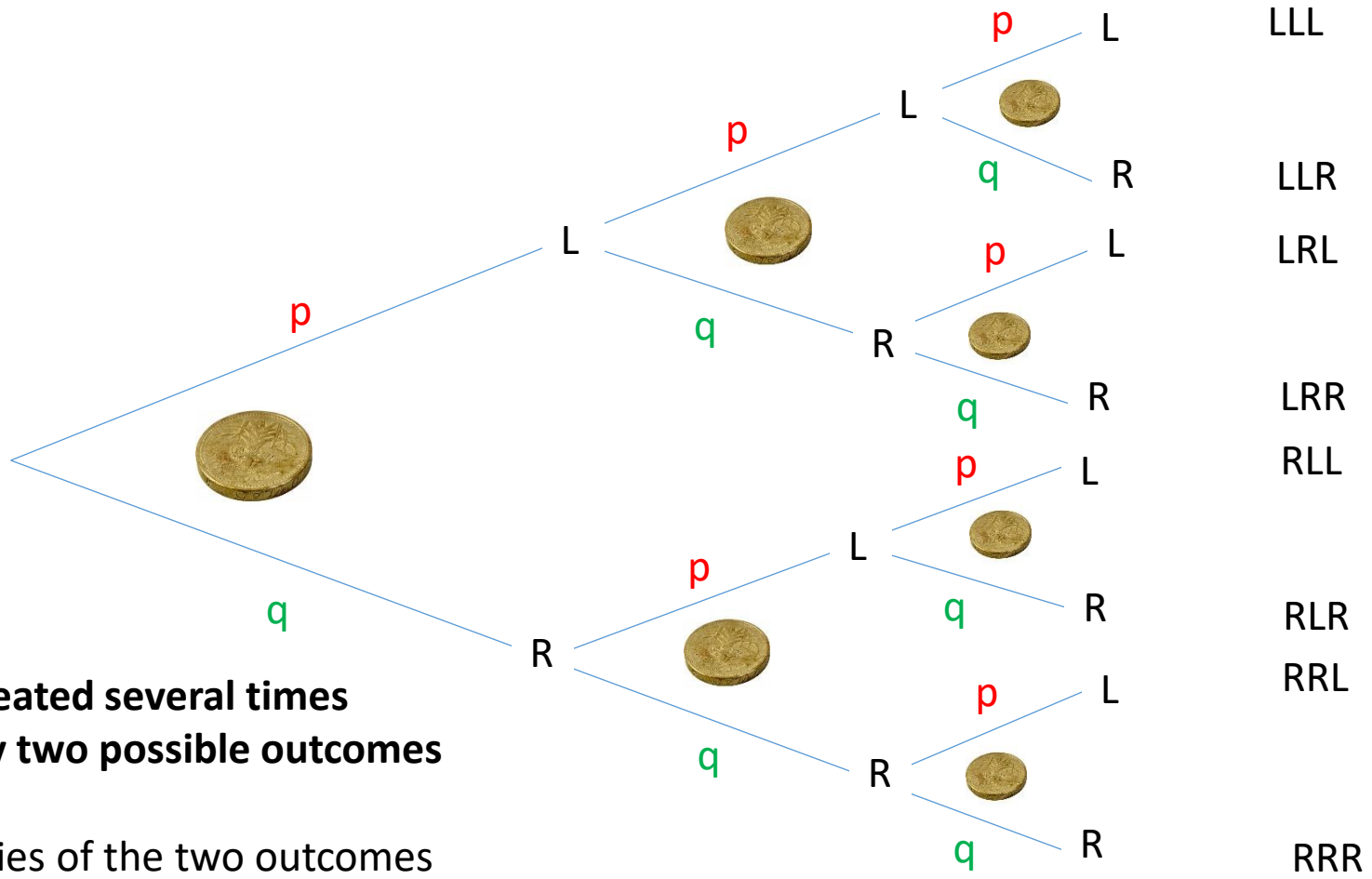
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Note that:

- Same experiment is **repeated several times**
- Each time there are **only two possible outcomes** (L or R)
- Each time the probabilities of the two outcomes are the same ($\boldsymbol{P(L) = p, P(R) = q}$)
- All the **repetitions are independent** of each other

Bernoulli Process & Bernoulli Trials

- An experiment that satisfies the conditions we saw in the previous example is called a **Bernoulli Process**, and each repetition is called a **Bernoulli Trial**.

In summary, a Bernoulli Process is a random process that satisfies these conditions

Bernoulli Process & Bernoulli Trials

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In summary, a Bernoulli Process is a random process that satisfies these conditions

1. It consists of **several repetitions**
2. Each repetition is **independent** of the other
3. Each repetition has **only two possible outcomes** (*usually called "**success**" and "**failure**"*)
4. The **probability of "success" is the same** during each repetition (*common notation* $\boldsymbol{p = P(\text{"success"}), q = P(\text{"}failure\text{"})}$)

Some Questions We Could Ask…

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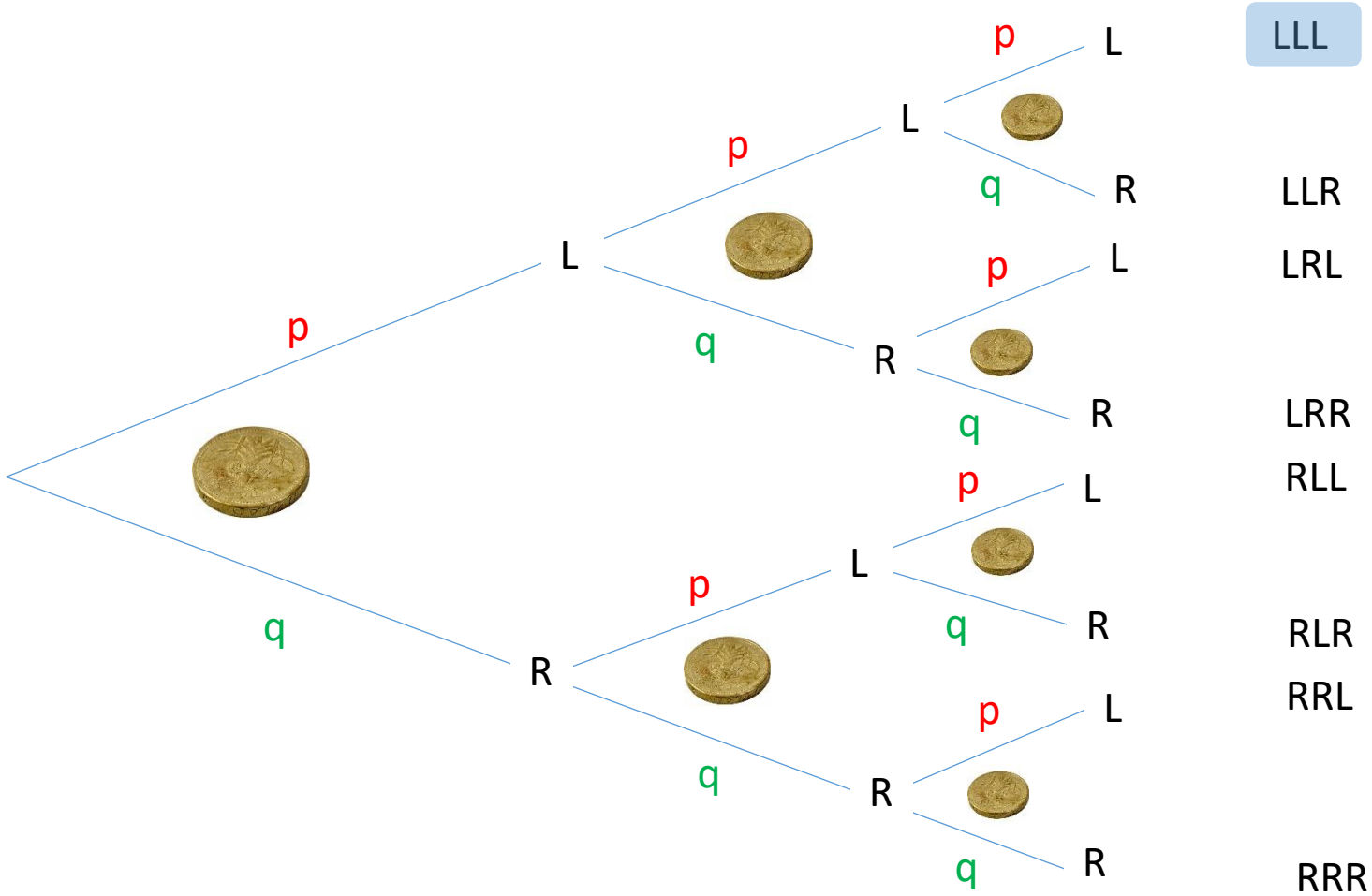
- What is the probability that in the n **repetitions** you take **exactly x left turns**?
- What is the probability that the **first left turn** you take is **at repetition number x**?
- What is the probability that it takes you x **repetitions to take exactly r left turns**?

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Each of these leads to a different random variable X, with its own probability mass function $f(x)$

What is the probability that you take three left turns?



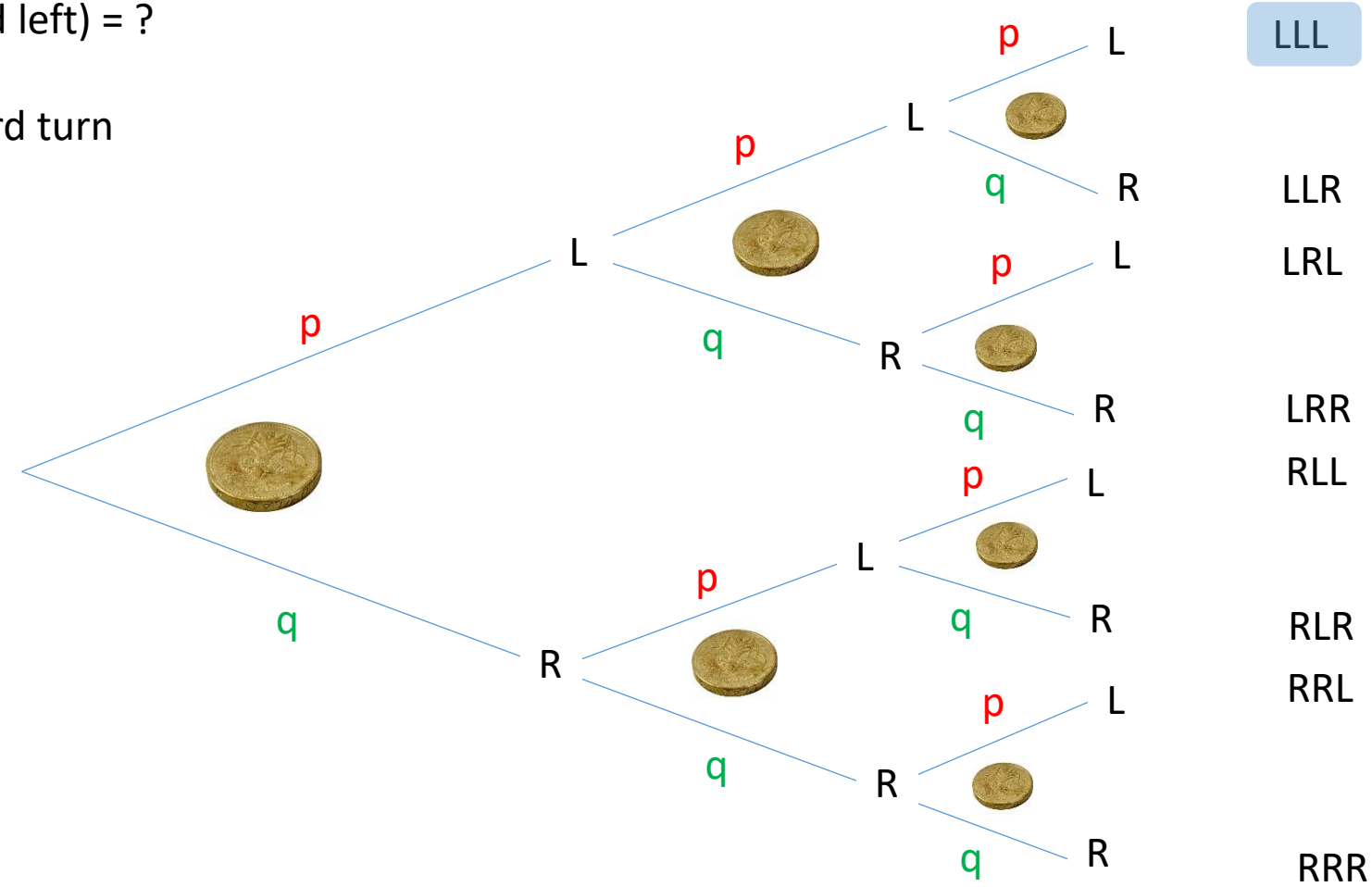
What is the probability that you take three left turns?

Probability (first left **and** second left **and** third left) = ?

Let $t_1 =$ first turn, $t_2 =$ second turn, $t_3 =$ third turn

Then,

$$P(t_1 = L \cap t_2 = L \cap t_3 = L) = ?$$



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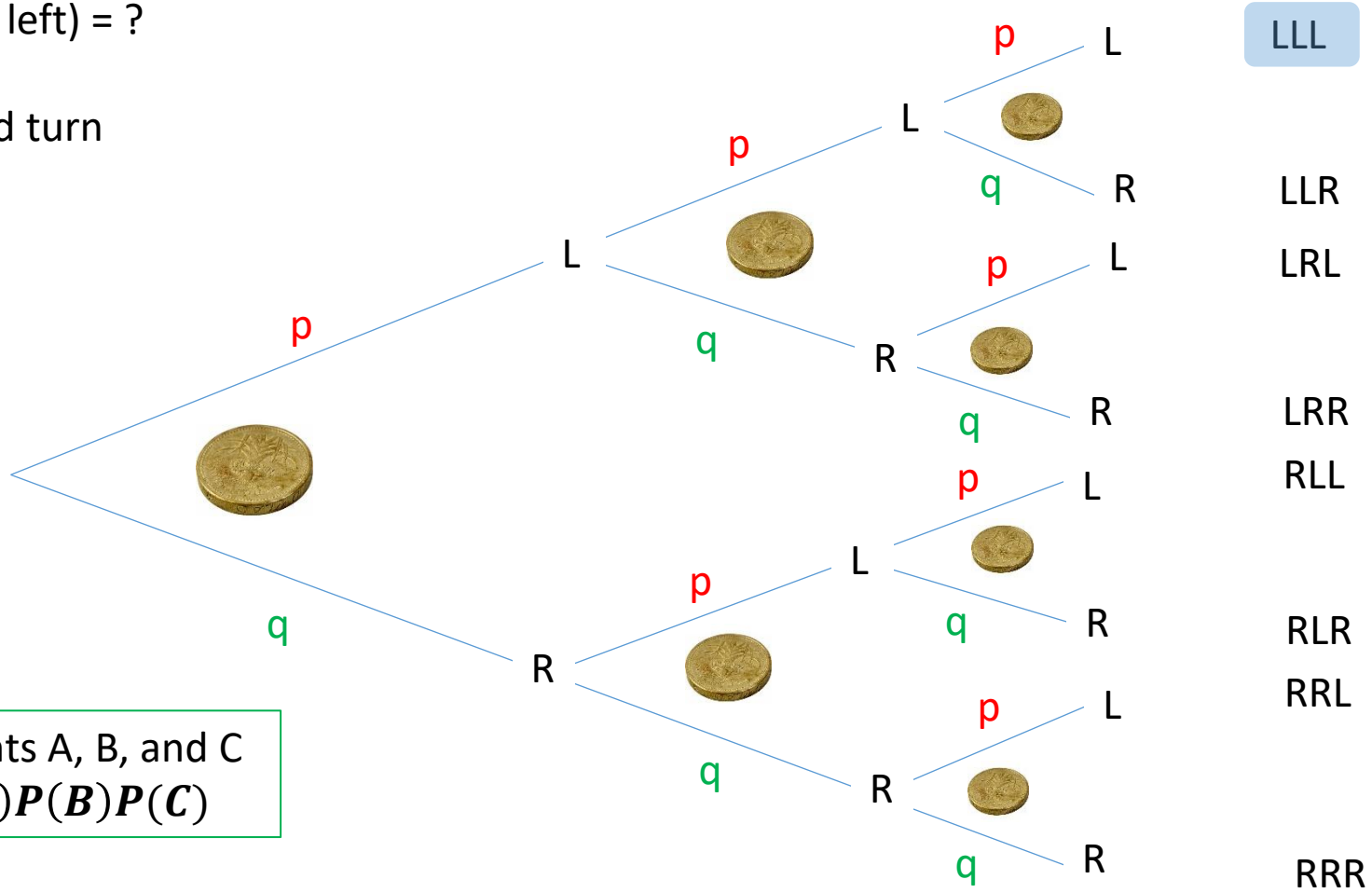
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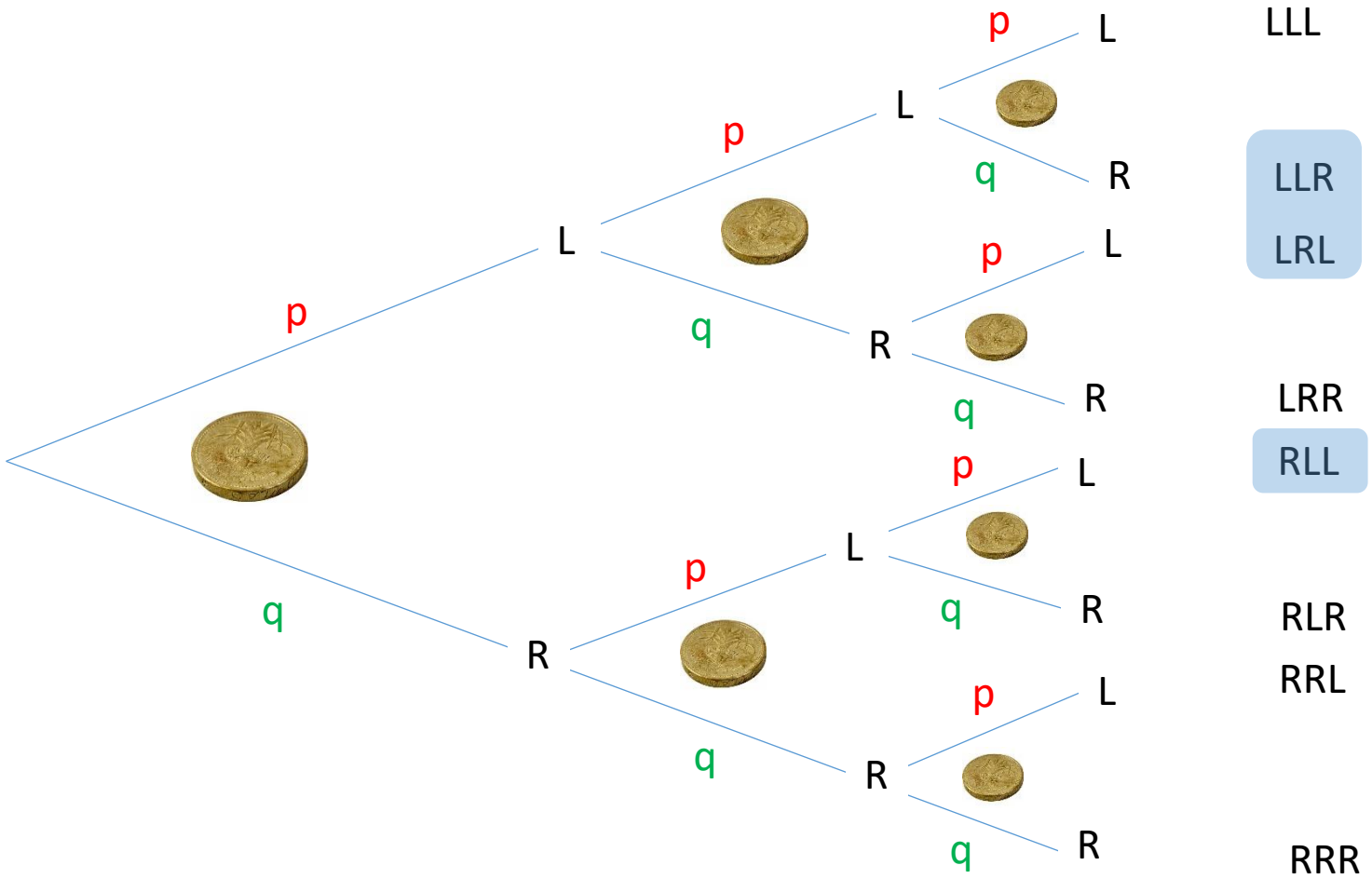
$$= P(t_1 = L)P(t_2 = L)P(t_3 = L)$$

$$= p \times p \times p = p^3$$

Recall that for **independent** events A, B, and C we have $\boldsymbol{P(A \cap B \cap C) = P(A)P(B)P(C)}$



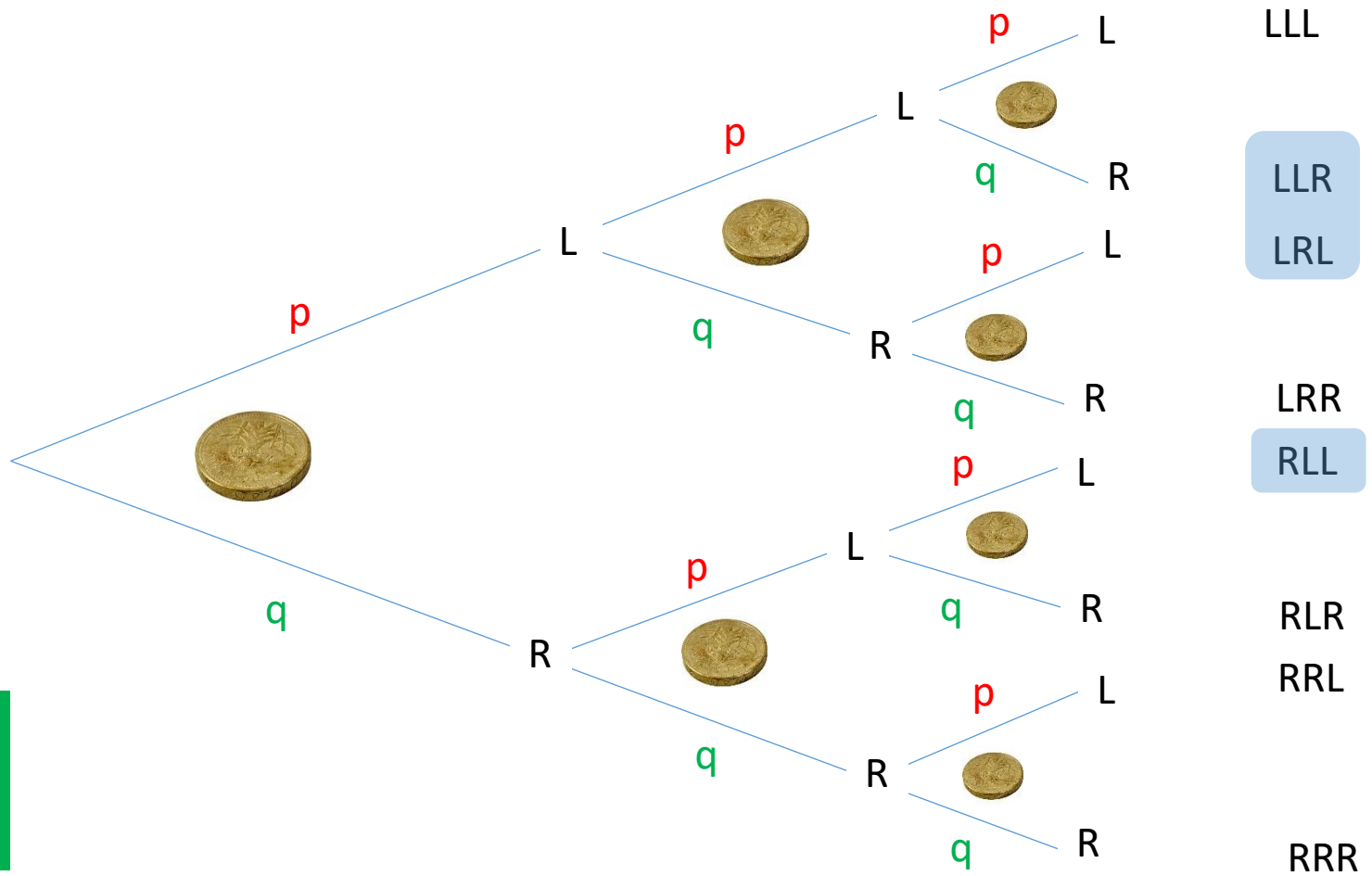
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What is the probability that you take exactly two left turns?

$$P(t_1 = L \cap t_2 = L \cap t_3 = R)$$
$$+P(t_1 = L \cap t_2 = R \cap t_3 = L)$$
$$+P(t_1 = R \cap t_2 = L \cap t_3 = L)$$

$$= ppq + pqp + qpp = 3p^2q$$



Next, We will use this understanding to answer several such questions, and develop various probability distributions

Binomial Distribution

Consider an experiment consisting of n Bernoulli trials (i.e., independent trials with constant probability of "success" denoted p)

Defining the setup (experiment)

Let $X = \boldsymbol{number\ of\ successes}\ in\ \boldsymbol{n}\ Bernoulli\ trials$

Defining the random variable

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Then the probability distribution of X is given by

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0, 1, \ldots, n$$

Finding formula for distribution of x

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And we say that X has a **Binomial Distribution**

Naming the distribution

Geometric Distribution

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$$f(x) = (1-p)^{x-1}p \qquad x = 1, 2, \ldots$$

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Negative Binomial Distribution

Consider an experiment consisting of Bernoulli trials (i.e., independent trials with constant probability of "success" denoted p)

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Let $X = \boldsymbol{number\ of\ trials}\ till\ \boldsymbol{r}\ successes$

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$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r \qquad x = r, r+1, r+2, \ldots$$

Finding formula for distribution of x

And we say that X has a **Negative Binomial Distribution**

Naming the distribution

Recall …

$$^{n}C_{r} \quad \text{or} \quad \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

Recall **5! = 5 x 4 x 3 x 2 x 1**
And **0! = 1**

Interpretations …

$X = \textbf{number of successes } in \textbf{ n } trials$

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0, 1, \ldots, n$$

Binomial Distribution

Interpretations …

$X = \boldsymbol{number\ of\ successes}\ in\ \boldsymbol{n}\ trials$

x successes

Failures = $n - x$

Total number of trials = n

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0, 1, \ldots, n$$

Binomial Distribution

$P(success) = p$

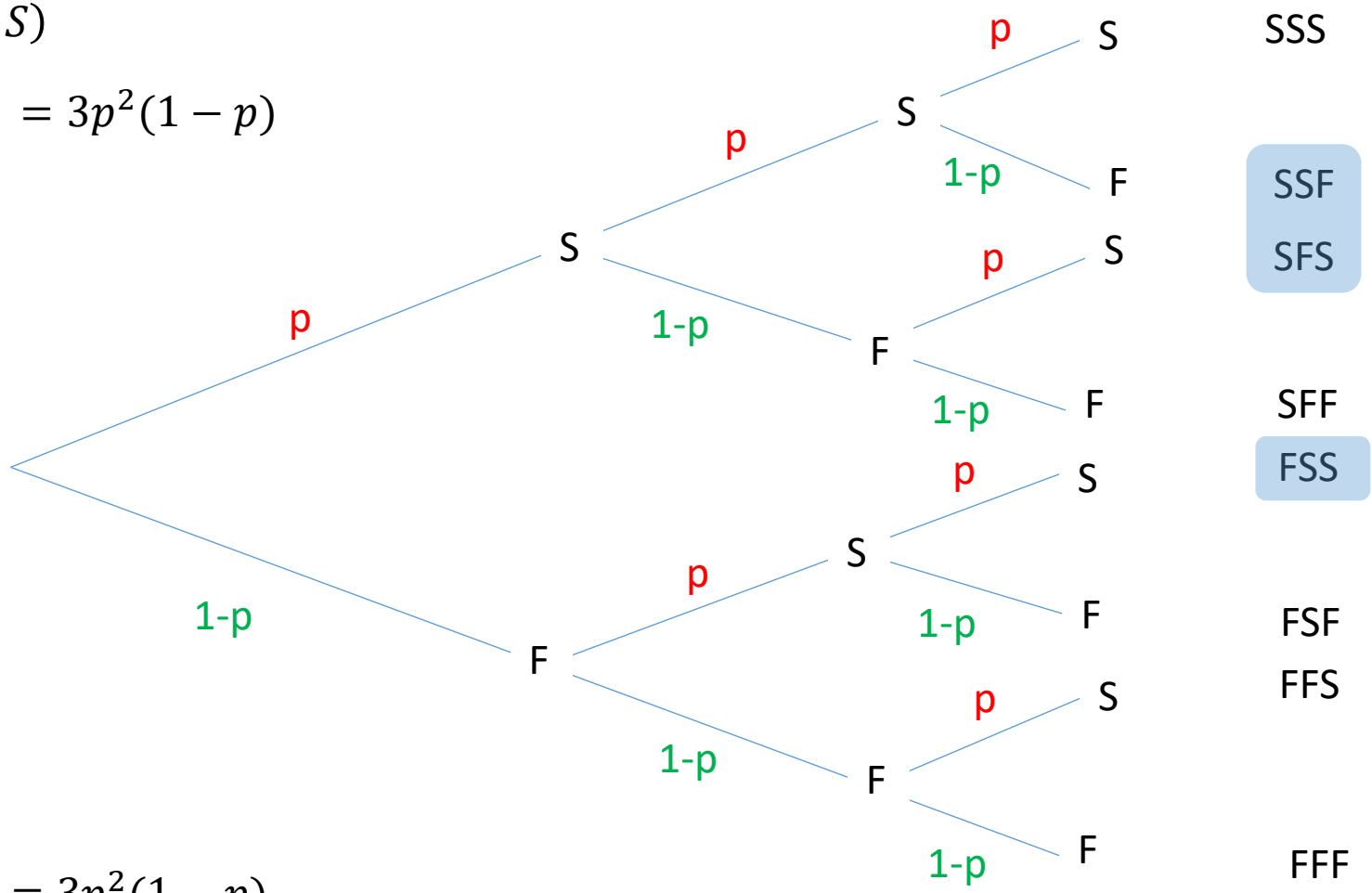
$P(failure) = 1 - p$

Number of ways we can have x successes in n trials knowing that the order does not matter

What is the probability of two success in three trials?

$$P(S \cap S \cap F) + P(S \cap F \cap S) + P(F \cap S \cap S)$$

$$= pp(1-p) + p(1-p)p + (1-p)pp \; = 3p^2(1-p)$$



Using the Binomial Distribution formula with $n = 3, x = 2$ we get the same result

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$f(2) = \binom{3}{2} p^2(1-p) = \frac{3!}{(3-2)!\,2!} p^2(1-p) = 3p^2(1-p)$$

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$X = \boldsymbol{number\ of\ trials}\ till\ first\ success$

$$f(x) = (1 - p)^{x-1}p \qquad x = 1, 2, \ldots$$

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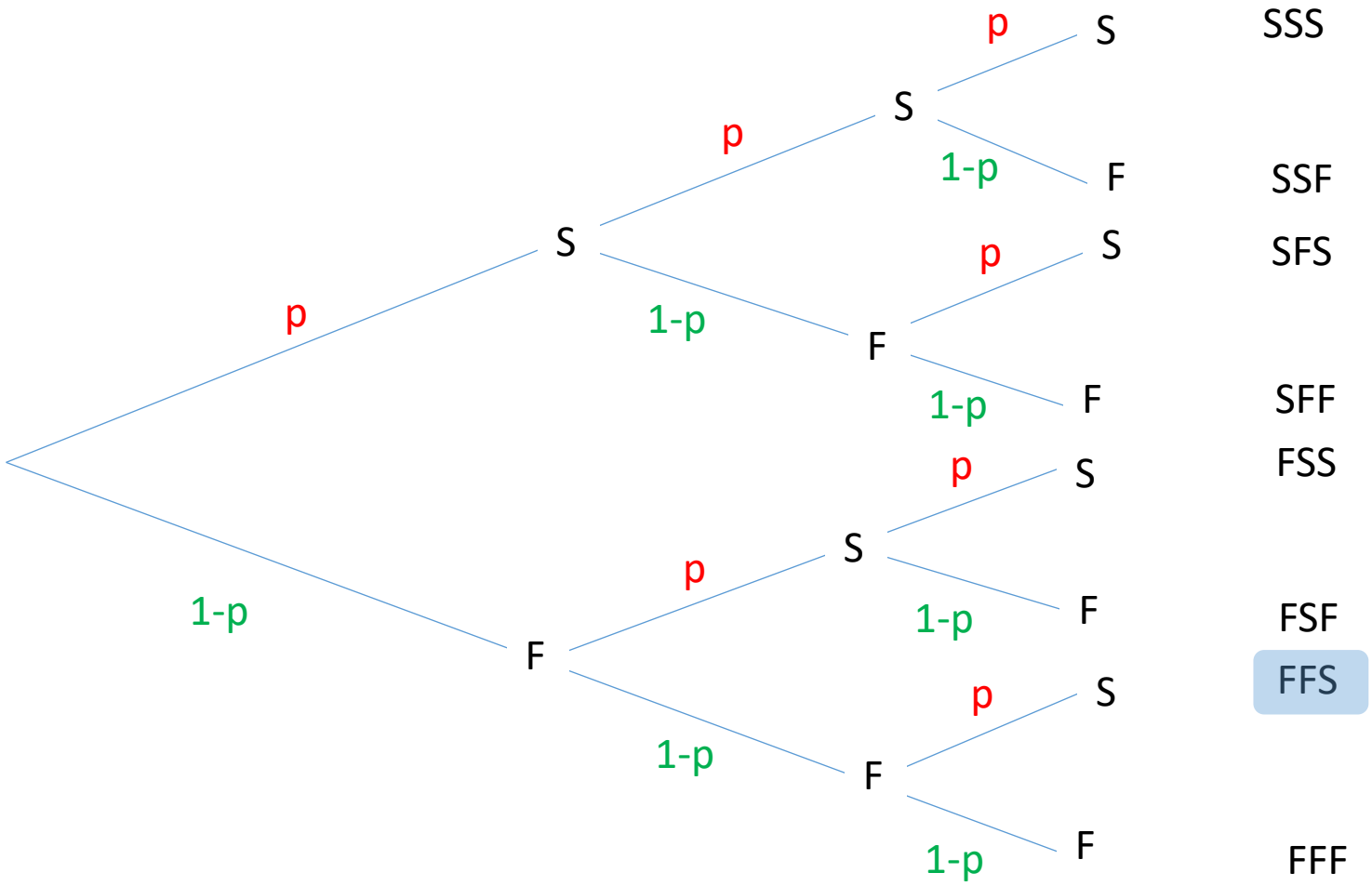
Success occurs in trial number x but not before

There are a total of $x - 1$ failures *before* the first success occurs

There has to be at least one trial for one success to occur (so no $x = 0$)

What is the probability of that first success occurs in third trial?

$$P(F \cap F \cap S)$$
$$= (1-p)(1-p)p = (1-p)^2 p$$



Using the Geometric Distribution formula with $x = 3$ we get the same result

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$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r \qquad x = r, r+1, r+2, \ldots$$

Negative Binomial Distribution

Interpretations …

$X = \textbf{number of trials } till \textbf{ r } successes$

There have to be at least r trials for r successes to occur

$x - r$ failures

r successes

$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^{r} \qquad x = r, r+1, r+2, \ldots$$

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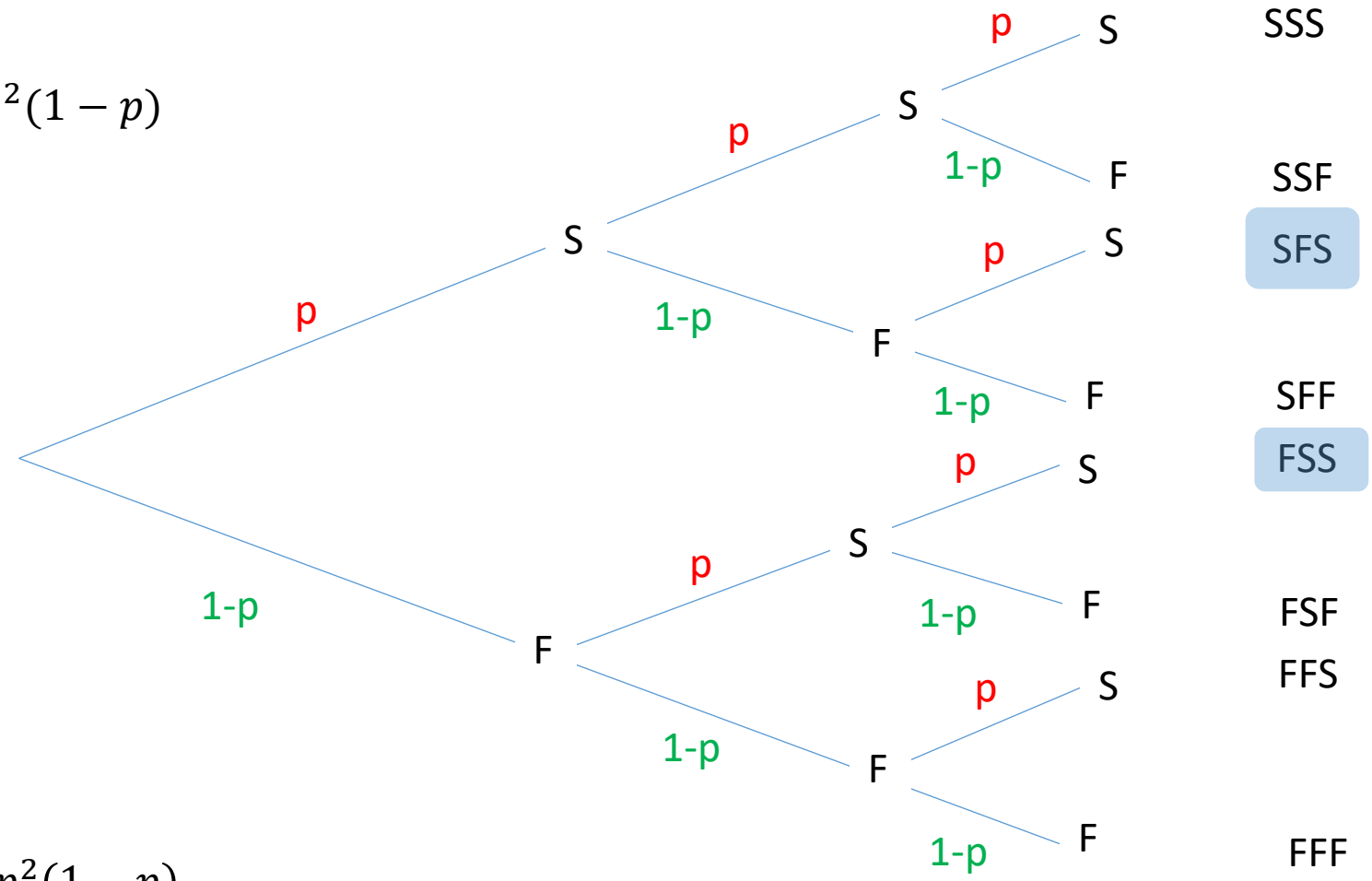
$P(success) = p$

By definition, we know that the last trial is a success, thus we only have to see in how many ways can we have remaining $r - 1$ successes in the remaining $x - 1$ trials, knowing that order does not matter in these $x - 1$ trials.

What is the probability that it takes three trials (not more or less) to get first two successes?

$$P(S \cap F \cap S) + P(F \cap S \cap S)$$

$$= p(1-p)p + (1-p)pp \; = 2p^2(1-p)$$



Using the Negative Binomial Distribution formula with $x = 3, r = 2$ we get the same result

$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r$$

$$f(3) = \binom{3-1}{2-1}p^2(1-p) = \frac{2!}{(2-1)!\,1!}p^2(1-p)$$

$$= 2p^2(1-p)$$

Summary of Properties (Range, Mean, Variance)

Binomial Distribution

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$x = 0, 1, \ldots, n$$

$$\mu_X = E[X] = np$$

$$\sigma_X^2 = V[X] = np(1-p)$$

Geometric Distribution

$$f(x) = (1-p)^{x-1} p$$

$$x = 1, 2, \ldots$$

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Negative Binomial Distribution

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Law of Rare Events

And the Poisson Distribution

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- But only a few actually do (i.e., each is very rare)

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Examples of large number of "rare" and "independent" events:

- Almost the whole world could call my number, but on a given day only a (relatively) "few" actually do!
- While all of the students in my class could fall asleep, only a few actually do (per lecture)!

Law of Rare Events

And the Poisson Distribution

Additional Examples

- Number of overflow floods in a river per decade
- Number of goals in a Soccer World Cup match
- Relatively large (diameter > 1 meter) meteorites hitting earth per year
- Machine malfunctions per year
- Locations of defects in materials

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Note that the “occurrences” do not have to be temporal (e.g., they could spatial as well).

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“Spatial” Occurrences: Number of chewing gums per tile on a sidewalk.

Law of Rare Events

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The question "how many of these rare independent events actually occur in a given interval" can often (i.e., with the additional conditions listed) be answered by the Poisson Distribution.

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Mathematically

Given $\lambda =$ Expected number of events per unit time (fixed, and $\lambda > 0$)

$$P[k \text{ events occur in interval } t] = p_X(k; \lambda t) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

$$k = 0, 1, 2, 3, \ldots$$

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$$k = 0, 1, 2, 3, \ldots$$

Non-negative integers (since it's the "number of events" whose likelihood we wish to know).

Poisson Distribution

A Closer Look

Keep in mind…

- $\lambda =$ Expected number of events per unit time (fixed, and $\lambda > 0$)
- $k = 0, 1, 2, 3, \ldots$

For unit time interval (i.e., $t = 1$), we may rewrite the Poisson Distribution as

$$p_X(k;\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

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- The distribution has only one parameter λ
- In fact, λ turns out to be both the mean and the variance of the distribution
  - i.e., $E[X] = \lambda, V[X] = \lambda$

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$p_X(6;10)$

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$$p_X(k;\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$p_X(6;10)$

- Probability that $X = 6$ given $\lambda = 10$.
- Probability that 6 events occur in unit time given that the expected number of events per unit time is 10.

- The distribution has only one parameter λ
- In fact, λ turns out to be both the mean and the variance of the distribution
  - i.e., $E[X] = \lambda, V[X] = \lambda$

Poisson Distribution

How Does it Look?

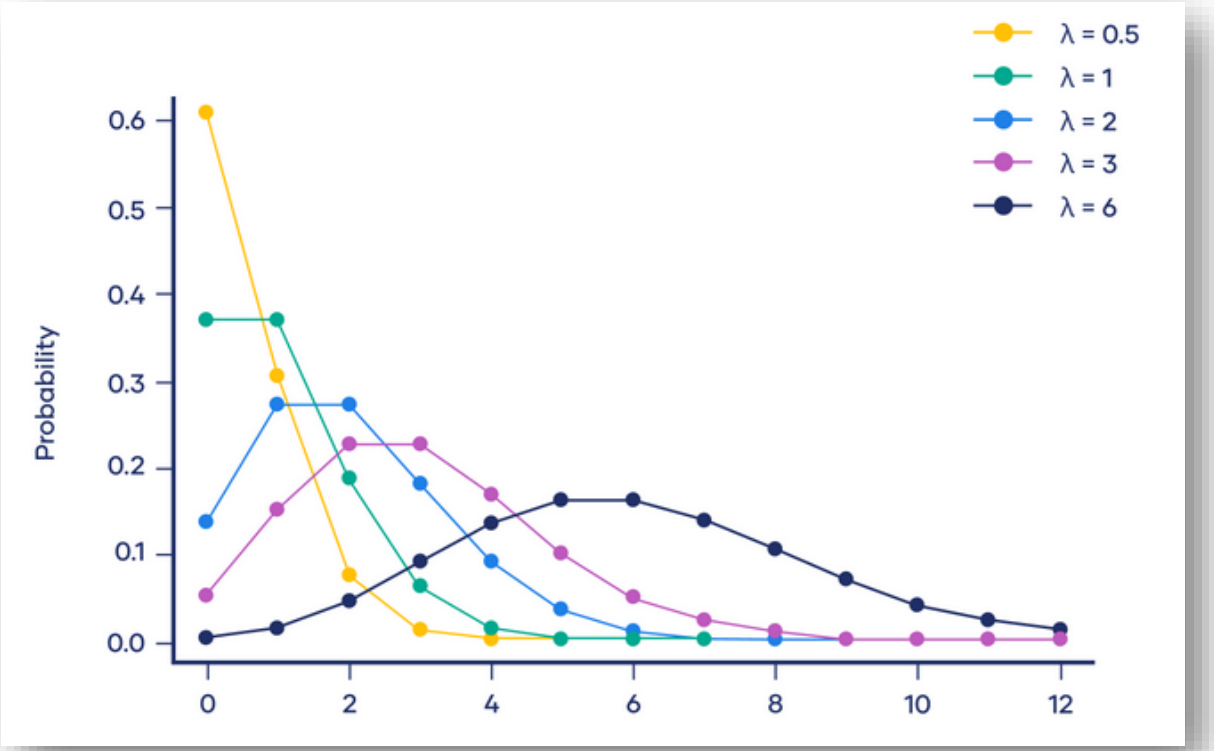
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Poisson Distribution

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Plotting for different values of λ

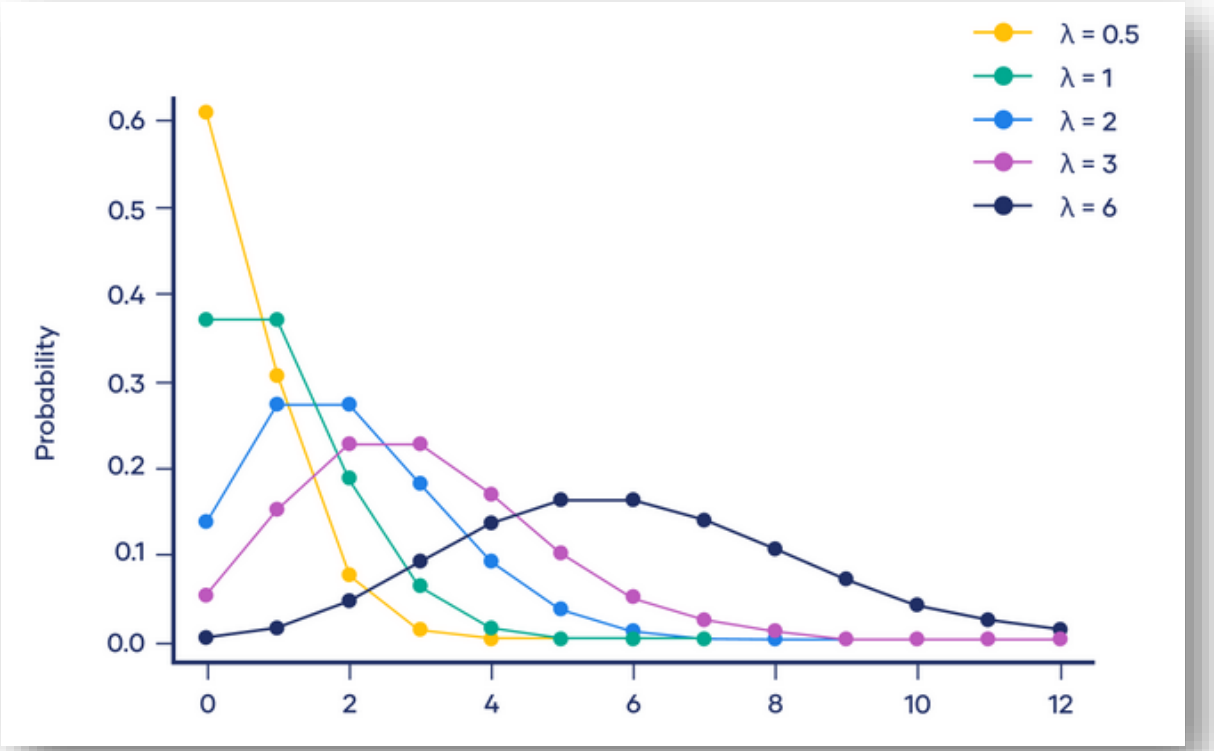


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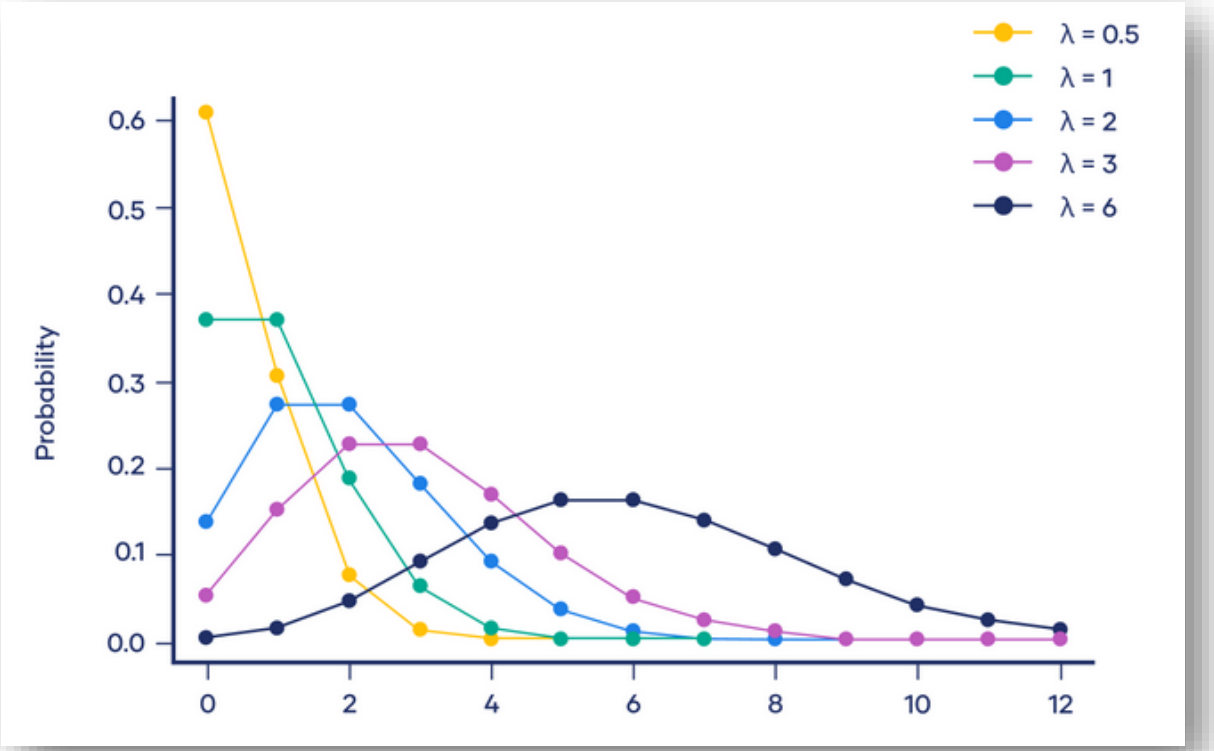
- Note that the probability peaks as we get close to the expected occurrence rate.
- So, for $\lambda = 6$, probability of observing 6 events in unit time is one of the highest.

Poisson Distribution

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Plotting for different values of λ



- Note that the probability peaks as we get close to the expected occurrence rate.
- So, for $\lambda = 6$, probability of observing 6 events in unit time is one of the highest.
- But as we move away from 6 the probability drops exponentially.
- Which makes sense, as e.g., probability of observing 2 events in unit time when fixed expected value is 6 should indeed be quite low.

Poisson Distribution

Example

$$p_X(k;\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

On a particular river, overflow floods occur once every 100 years on average. Calculate the probability of k = 0, 1, 2, 3, 4, 5, or 6 overflow floods in a 100-year interval, assuming the Poisson model is appropriate.

Poisson Distribution

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On a particular river, overflow floods occur once every 100 years on average. Calculate the probability of k = 0, 1, 2, 3, 4, 5, or 6 overflow floods in a 100-year interval, assuming the Poisson model is appropriate.

Because the average event rate is one overflow flood per 100 years, $\lambda = 1$

$$P(k \text{ overflow floods in 100 years}) = \frac{\lambda^k e^{-\lambda}}{k!} = \frac{1^k e^{-1}}{k!}$$

$$P(k = 0 \text{ overflow floods in 100 years}) = \frac{1^0 e^{-1}}{0!} = \frac{e^{-1}}{1} \approx 0.368$$

$$P(k = 1 \text{ overflow flood in 100 years}) = \frac{1^1 e^{-1}}{1!} = \frac{e^{-1}}{1} \approx 0.368$$

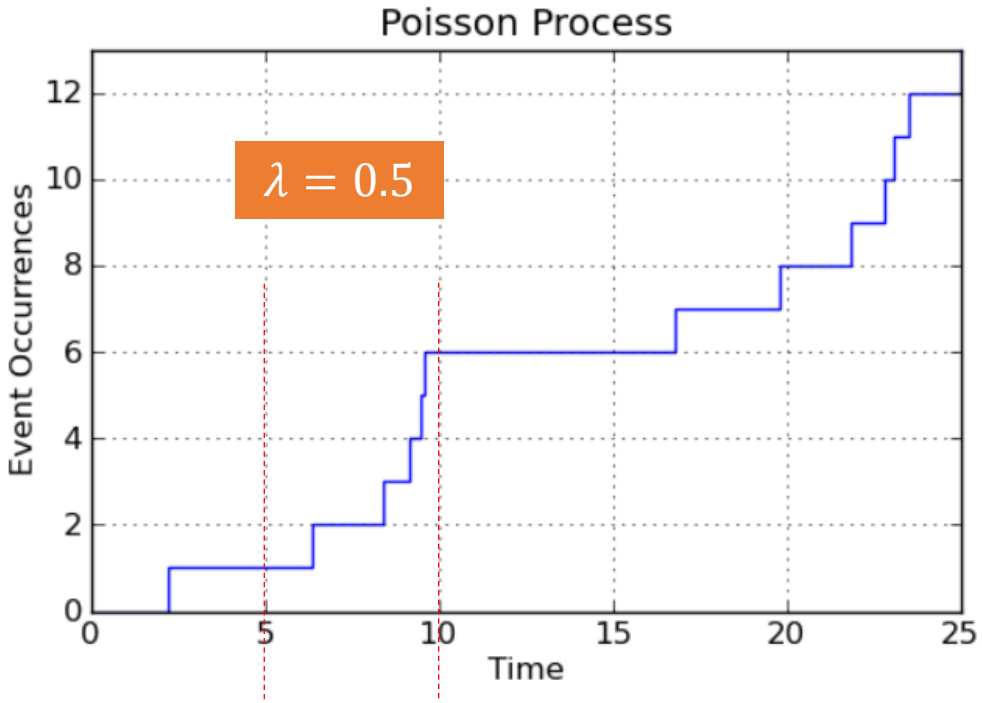
$$P(k = 2 \text{ overflow floods in 100 years}) = \frac{1^2 e^{-1}}{2!} = \frac{e^{-1}}{2} \approx 0.184$$

Poisson Process

Keeping Count of Poisson Distributed Events

Poisson Process

Keeping Count of Poisson Distributed Events

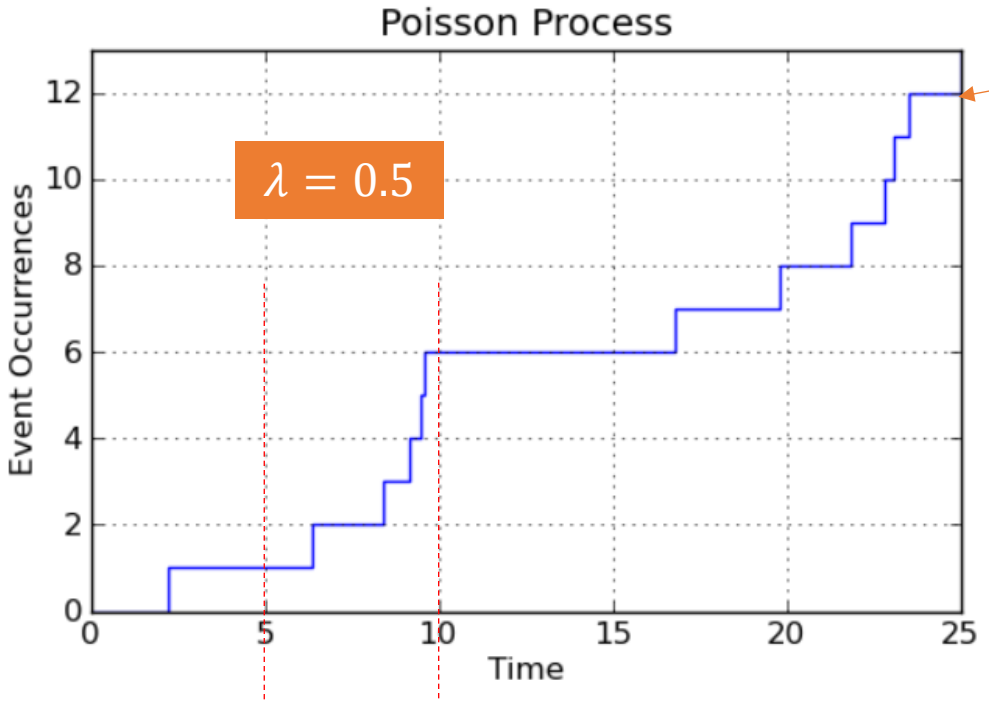


- Y-axis represents a count of how many Poisson distributed events have occurred in interval $(0\ t]$.

- In addition, number of events in any finite interval, for instance $(5\ 10]$, are also Poisson distributed.

Poisson Process

Keeping Count of Poisson Distributed Events



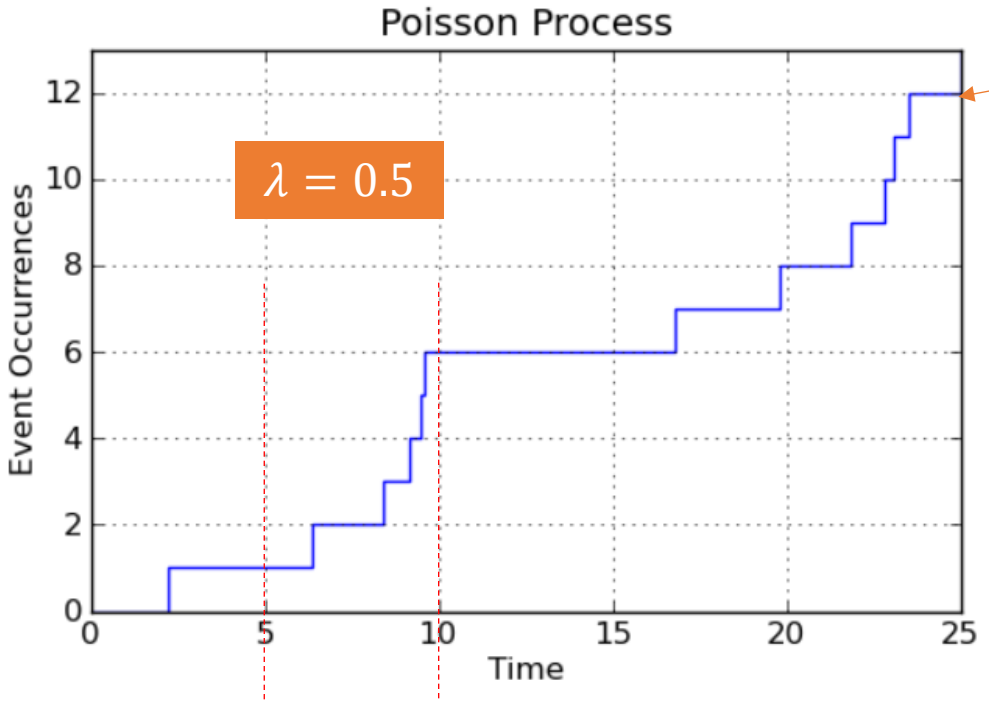
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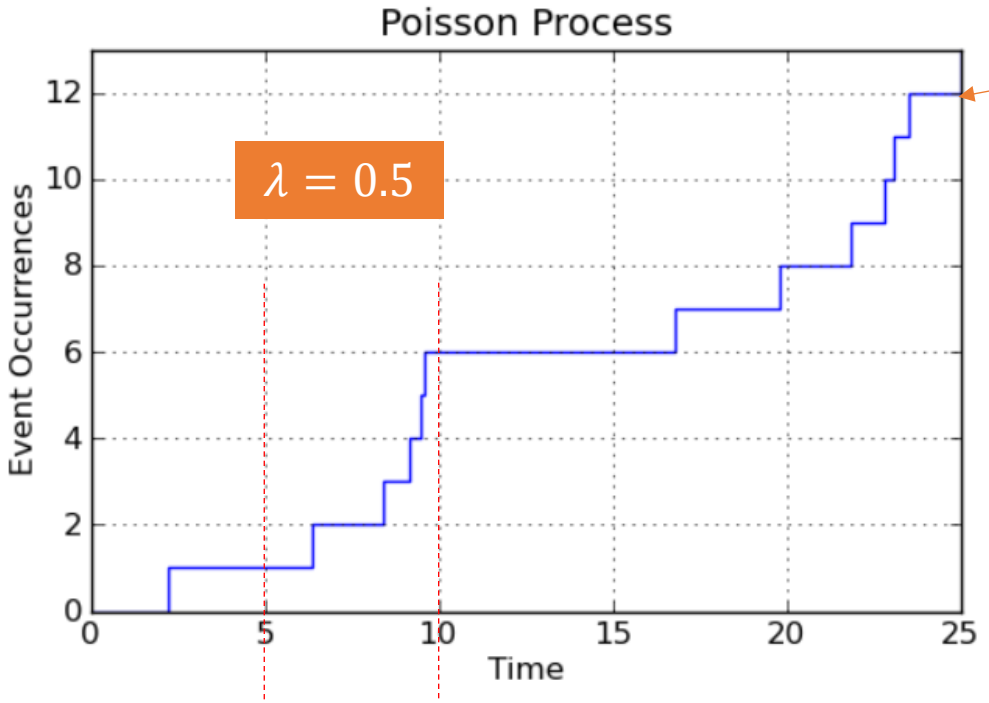
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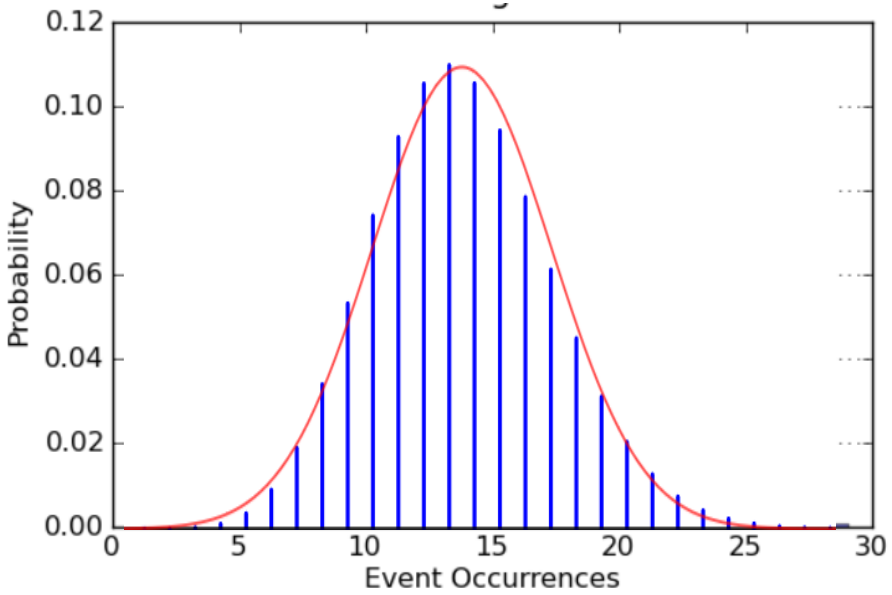
Keeping Count of Poisson Distributed Events



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$$p_{Y_{25}}(k;\lambda t) = \frac{(0.5 \times 25)^k e^{-(0.5\times 25)}}{k!}$$

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Poisson Distribution

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Poisson Distribution

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Do not use Poisson if

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Btw…

- We do have variants of Poisson Distribution that relax some of the assumptions.

- HW: read up on “Mixed Poisson ” , “Compound Poisson”, and “Zero-Truncated Poisson”

Where Does the Poisson Distribution Formula Come From?

Poisson Limit Theorem (PLT)

Another name for Law of Rare Events!

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Recall: Binomial Distribution

- If there are n events that can occur independent of each other such that
  - Each one either occurs with probability p, or does not occur with probability $1 - p$
  - Then the probability that k of them will occur follows the Binomial Distribution

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Poisson Distribution is a limiting case of the Binomial Distribution!

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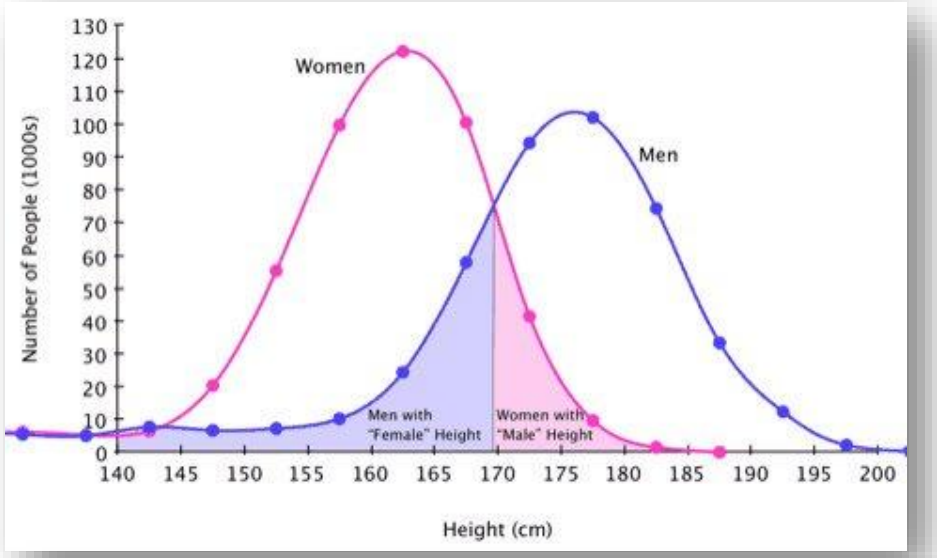
The Gaussian Distribution (aka Normal Distribution)

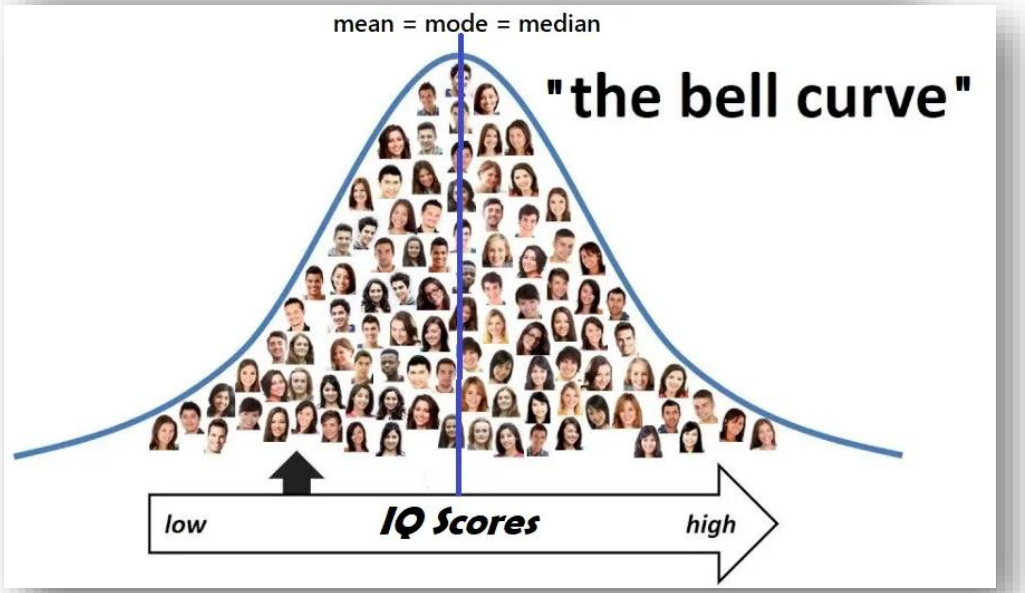
…and the **Central Limit Theorem**

Gaussian Distribution

And the Central Limit Theorem (CLT)

We are all familiar with the “bell curve”.



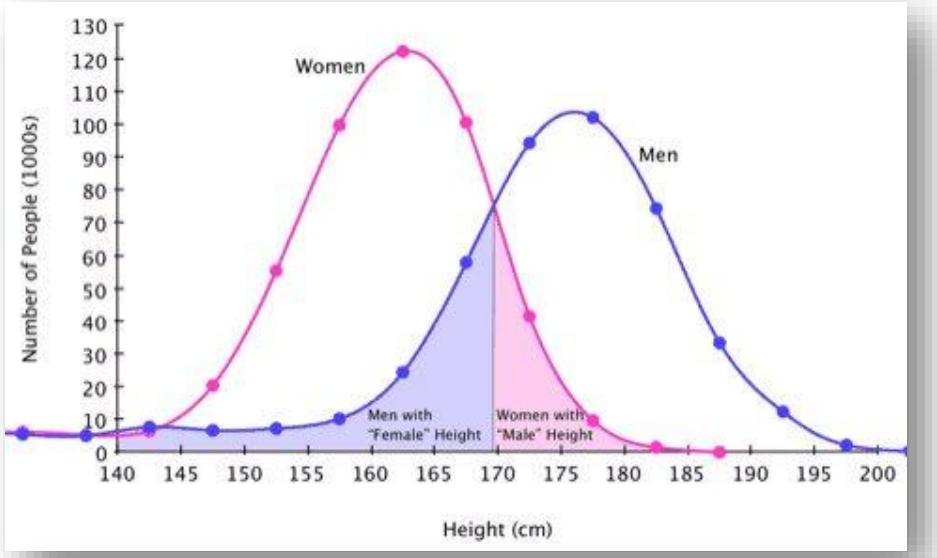


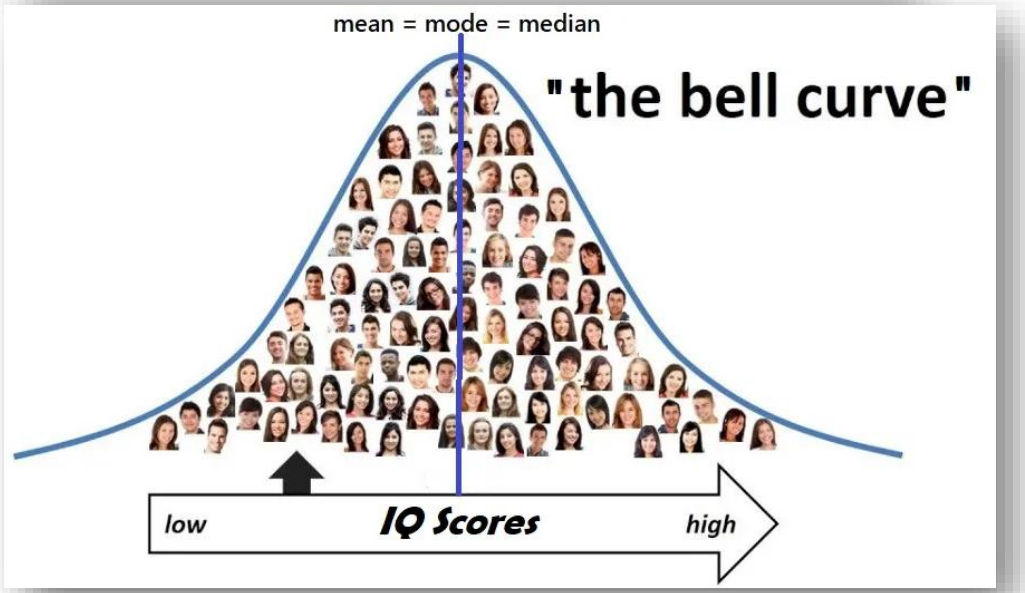
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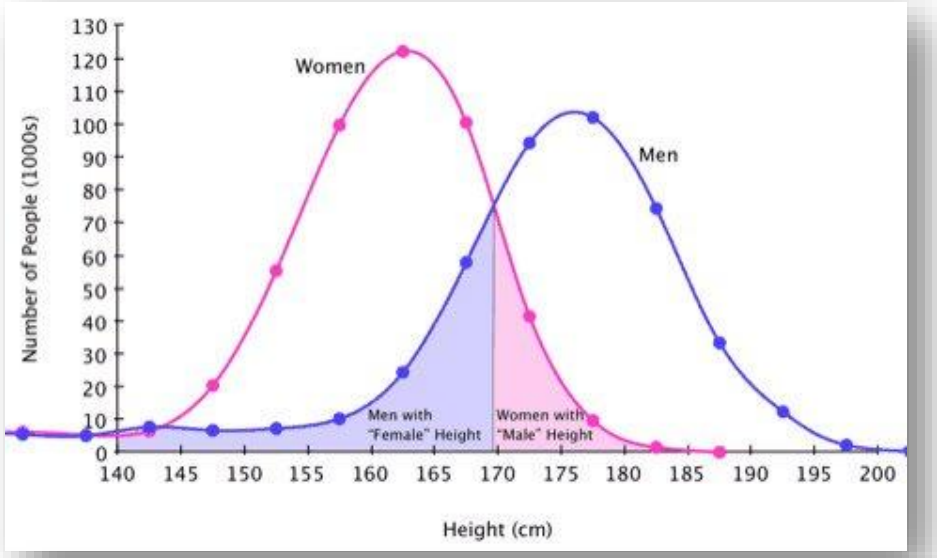
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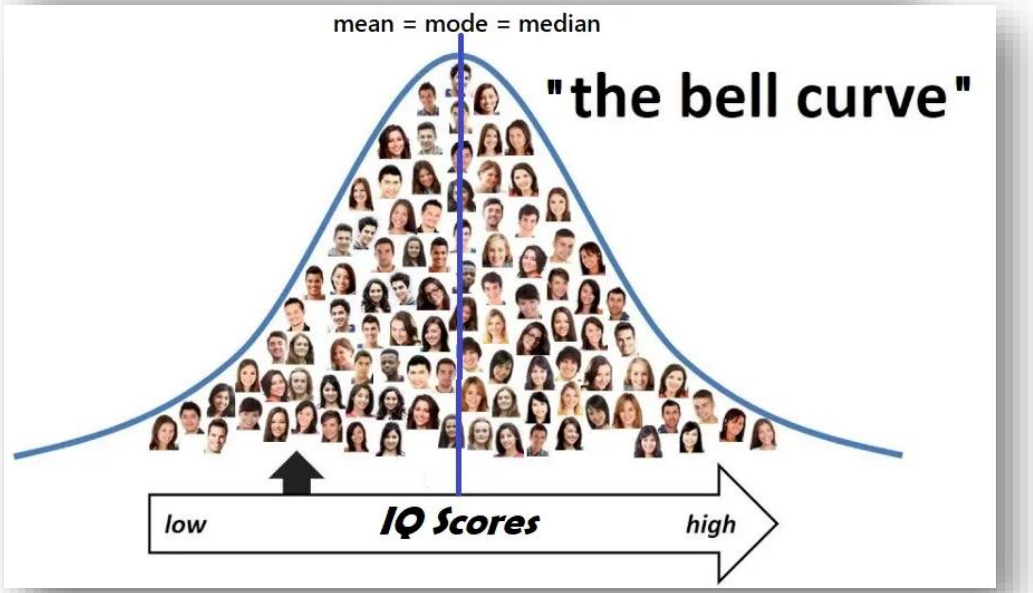
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    - E.g., think of heights (or weights) of people, typical measurement errors in experiments etc.





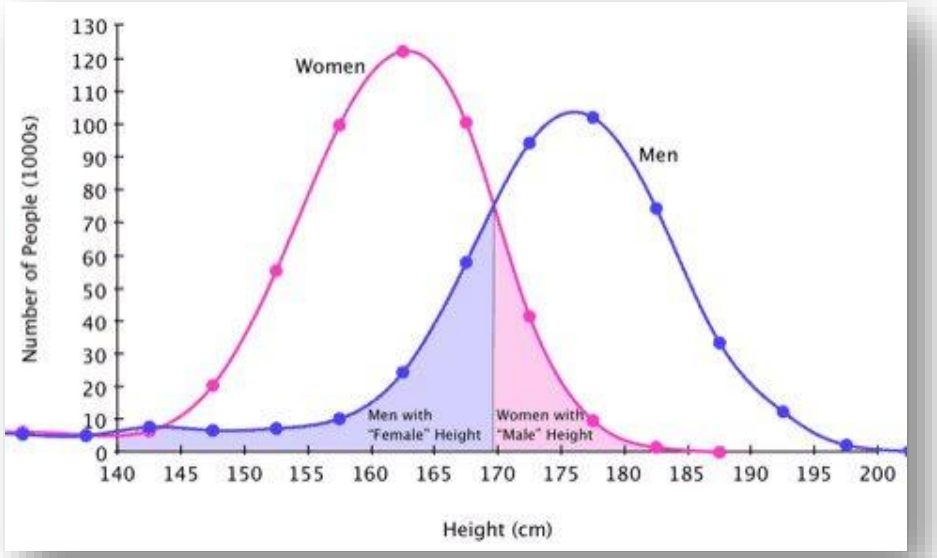
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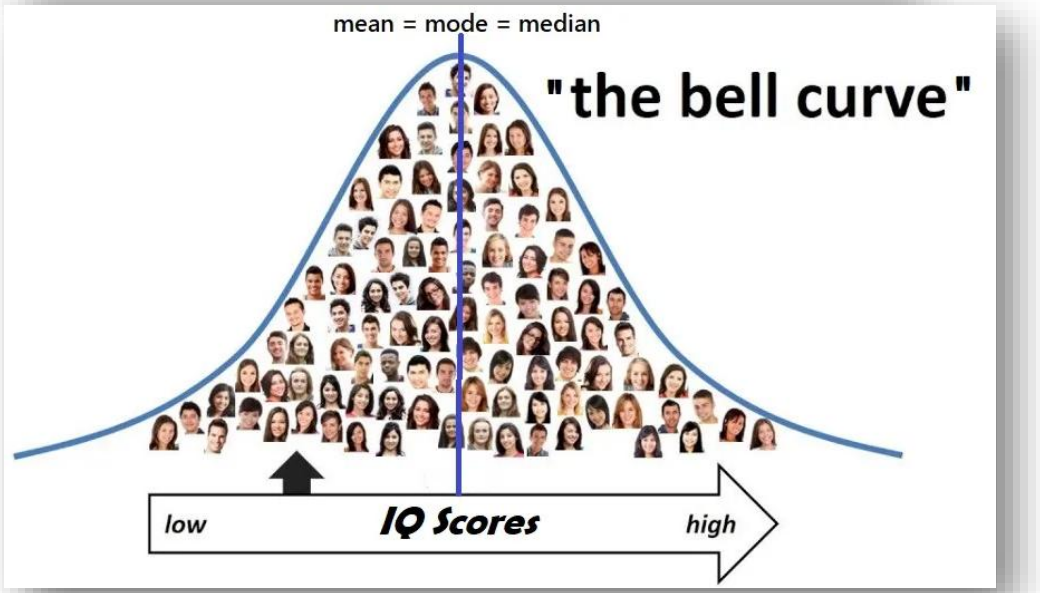
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  - E.g., think of heights (or weights) of people, typical measurement errors in experiments etc.
- Many random phenomena we observe are in fact an overall/cumulative (or “average”) effect of several underlying random occurrences.
  - CLT tells us that random phenomena thus created often follow a special distribution that we call the Gaussian/Normal Distribution.





Central Limit Theorem in Action

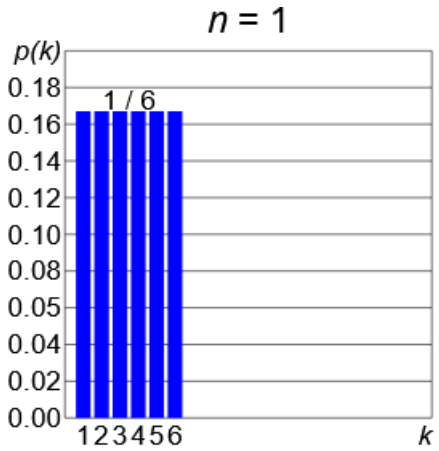
Let us consider a die roll experiment, where you win same amount of Rs. as the number showing on each die roll.

$$X_n = overall\ amount\ you\ win\ after\ n\ rolls$$

Let us look at the distribution of X_n for $n = 1, 2, 3, \ldots$

Central Limit Theorem in Action

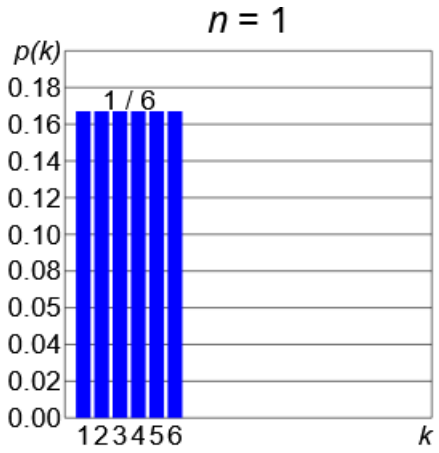
After one roll



(distribution of X_1)

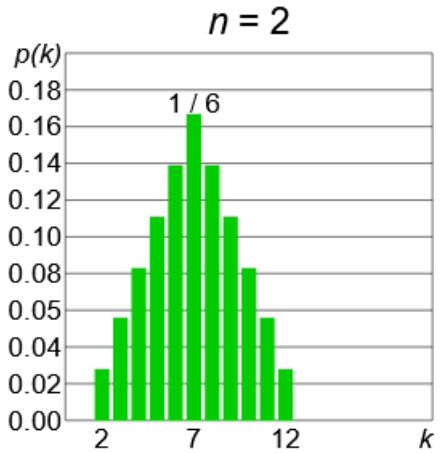
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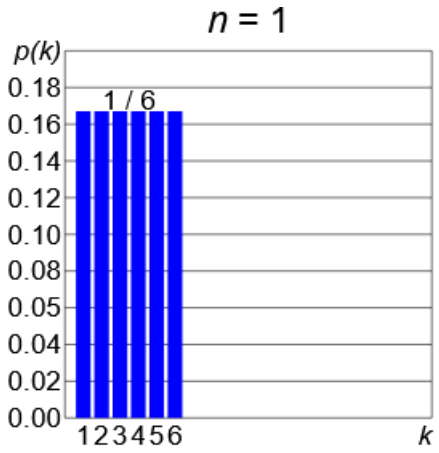
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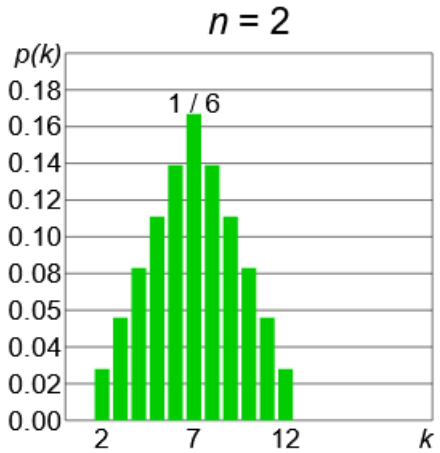
Central Limit Theorem in Action

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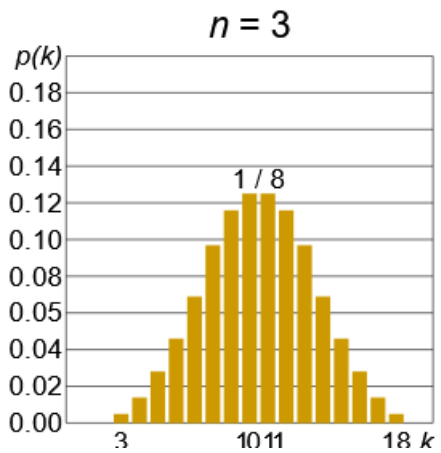
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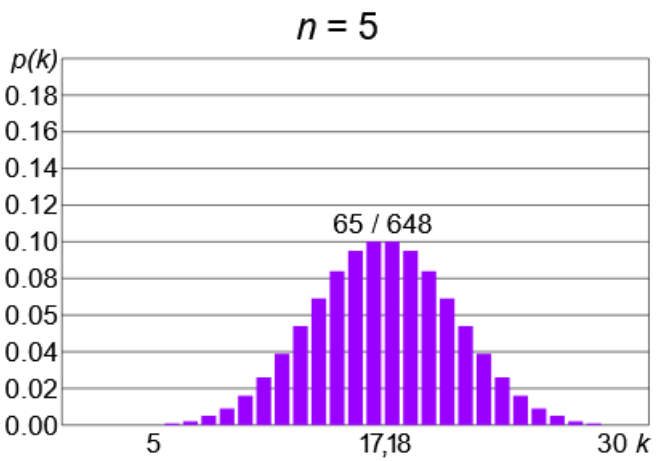
(distribution of X_2)

After three rolls



(distribution of X_3)

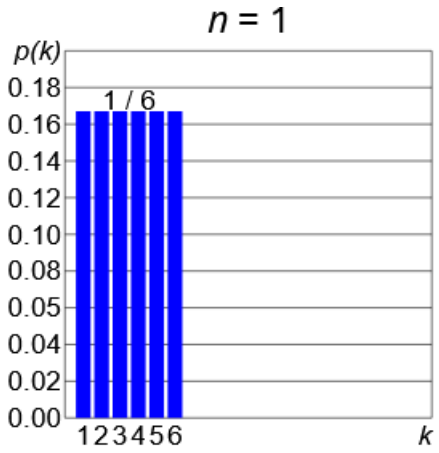
After five rolls



(distribution of X_5)

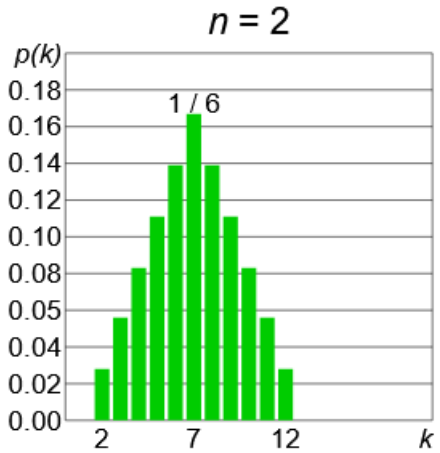
Central Limit Theorem in Action

After one roll



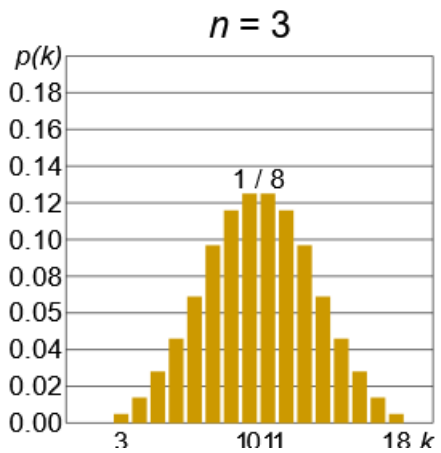
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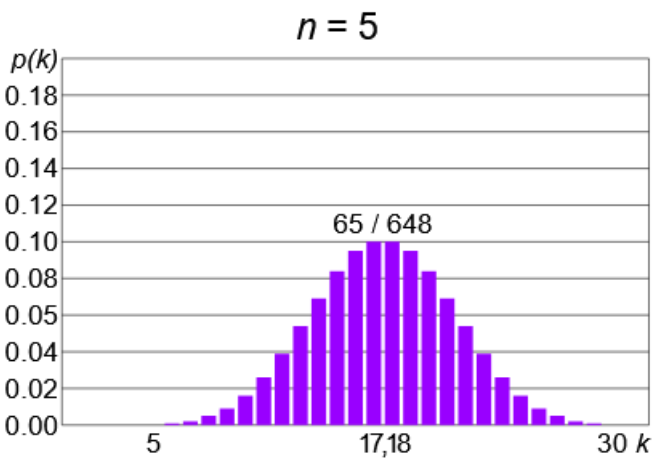
(distribution of X_2)

After three rolls



(distribution of X_3)

After five rolls



(distribution of X_5)

Notice how fast the cumulative effect of die rolls approaches the bell curve (even though each individual die roll follows a uniform distribution!)

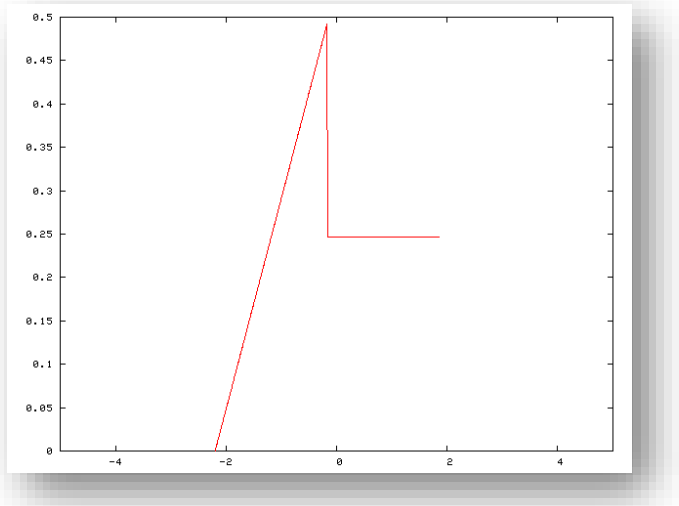
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Let V_i be independent random variables, each with the rather odd-looking distribution shown below

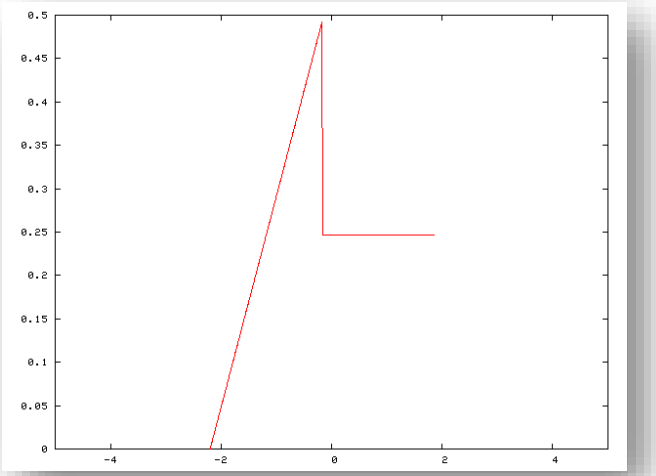


Distribution of each V_i

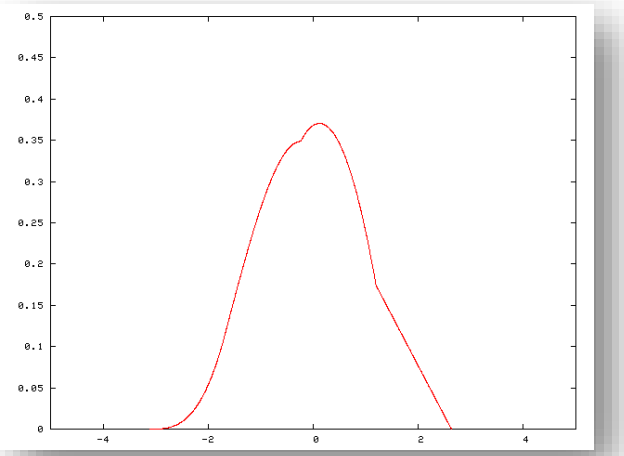
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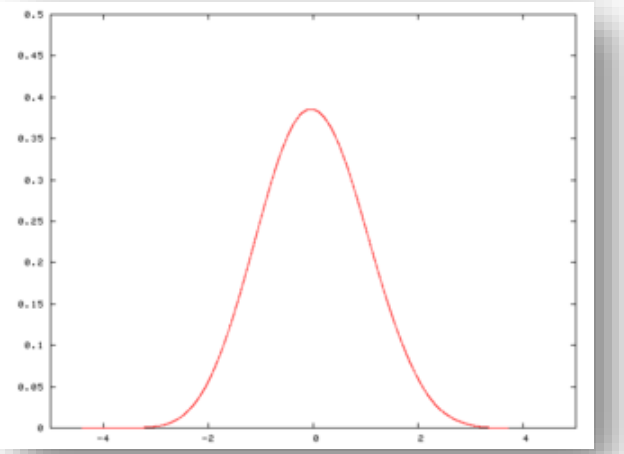
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Distribution of $V_1 + V_2$

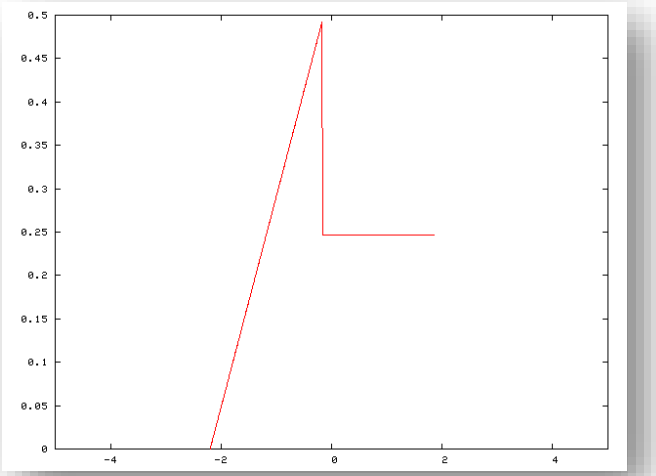


Distribution of $V_1 + V_2 + V_3 + V_4$

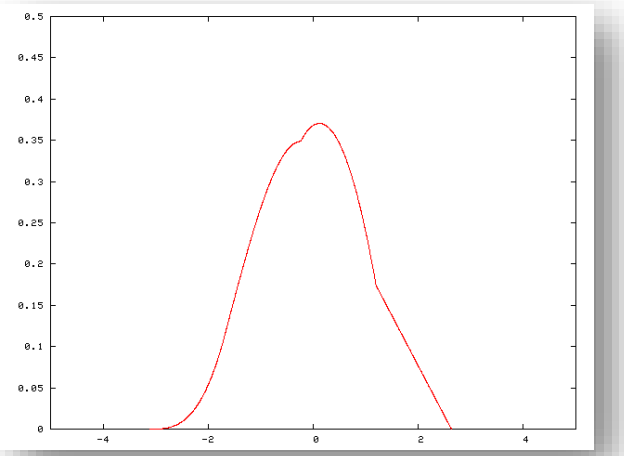
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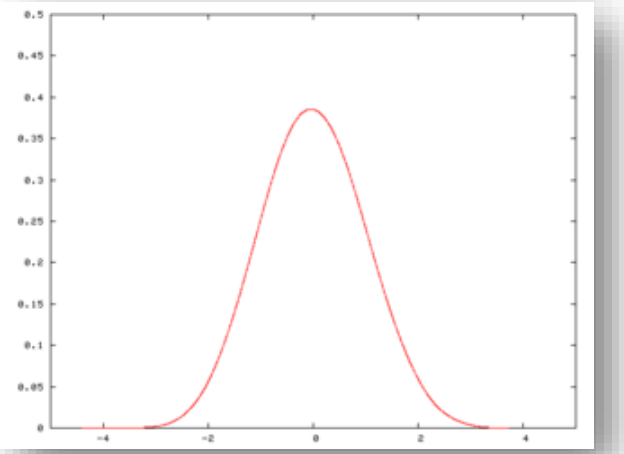
Let V_i be independent random variables, each with the rather odd-looking distribution shown below



Distribution of each V_i



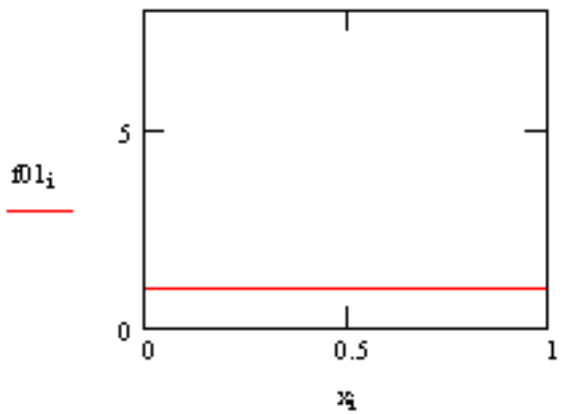
Distribution of $V_1 + V_2$



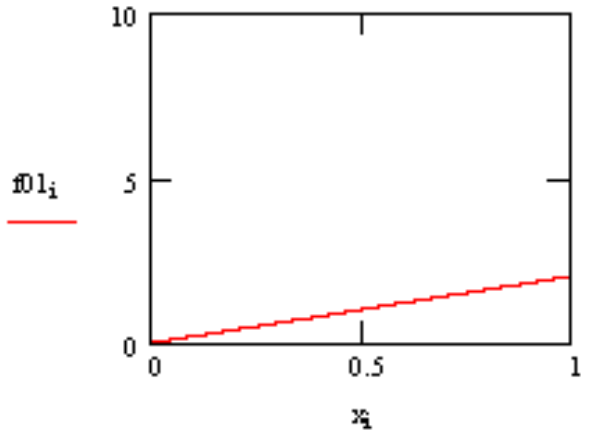
Distribution of $V_1 + V_2 + V_3 + V_4$

Notice again, how quickly the distribution of the sum (combined effect) approaches the bell curve!

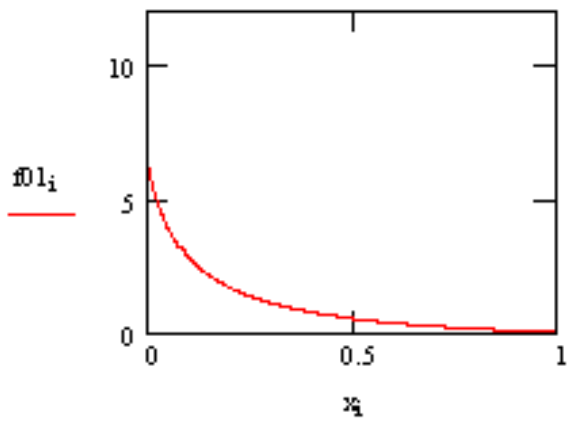
Some more examples…



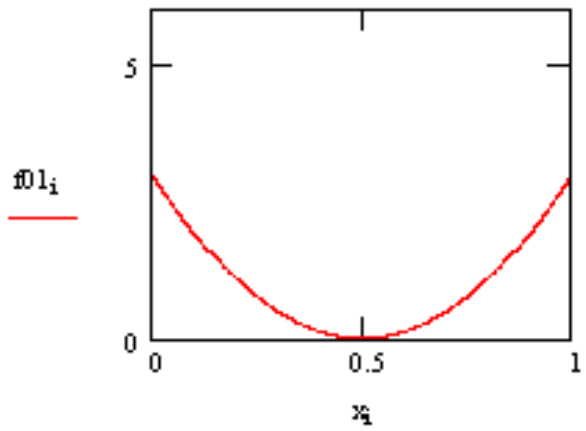
NonNormal Distribution of X



NonNormal Distribution of X



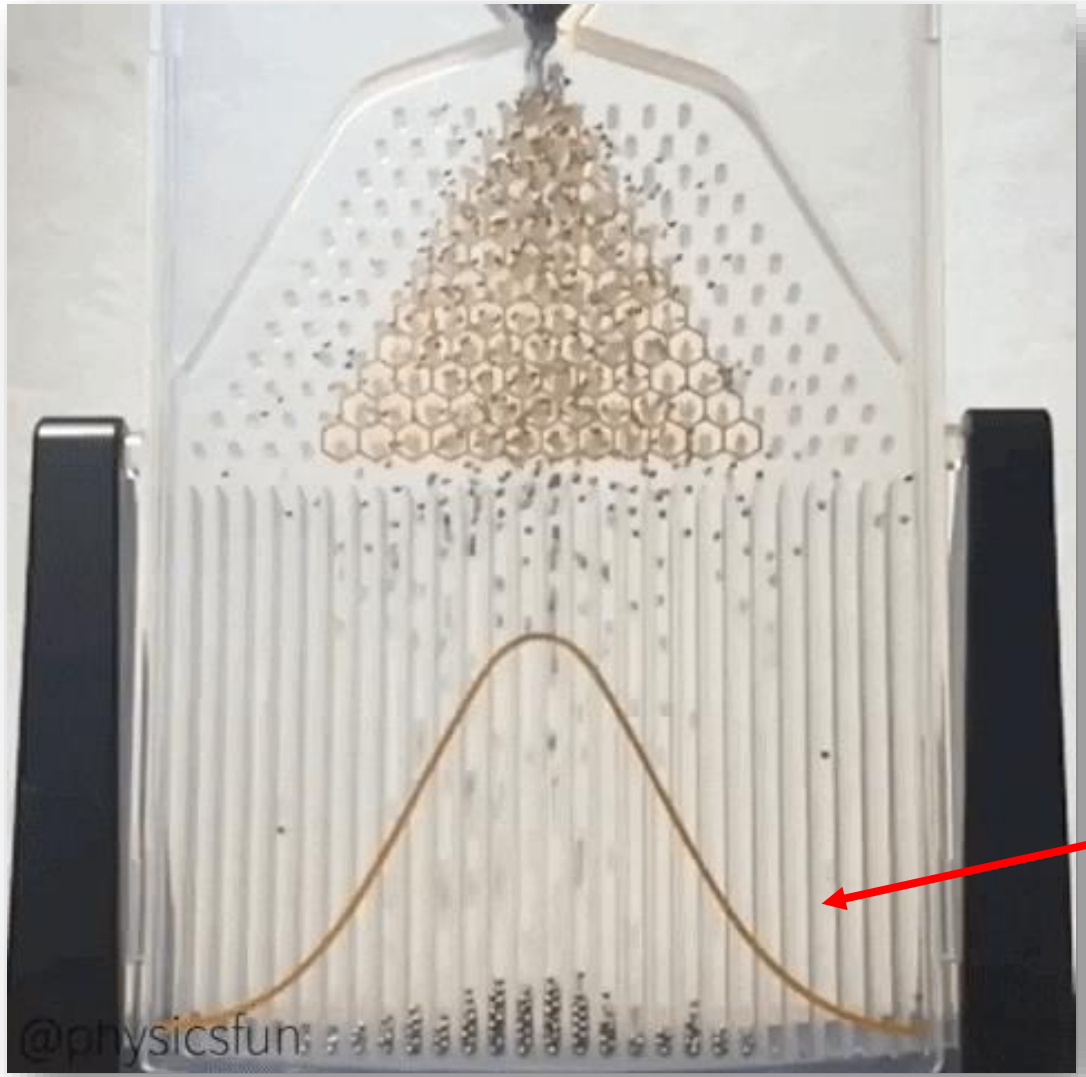
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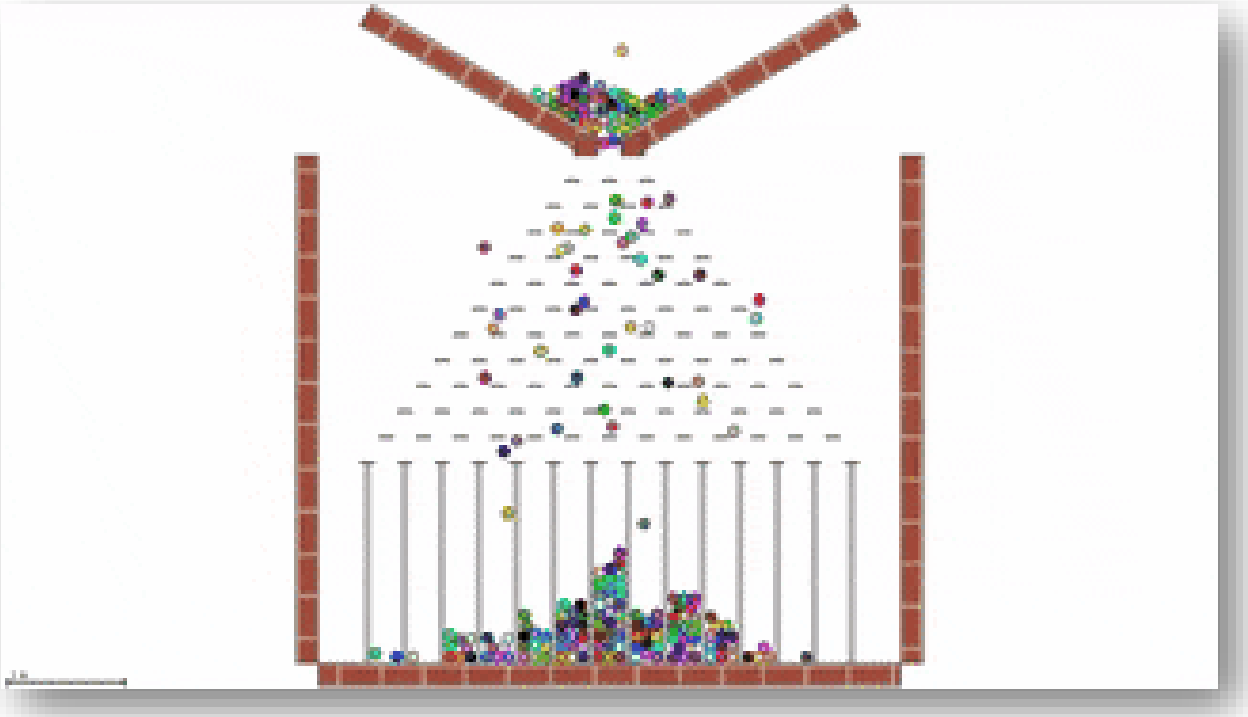


NonNormal Distribution of X

Central Limit Theorem in Action

Galton Board





Cumulative effect of several underlying random occurrences (balls going left or right at each peg).

Central Limit Theorem

…in simple words

Basic CLT

The average [behavior] of iid random variables follows a Gaussian Distribution.

Central Limit Theorem

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"iid"?

- Random variables $X_1, X_2, X_3, \ldots$ are said to be independent and identically distributed (iid) if
  - They are all independent of each other.
  - Each of them follows the exact same distribution.
    - E.g., each one is Gaussian with same mean and variance.

Central Limit Theorem

…in simple words

Basic CLT

The average [behavior] of iid random variables follows a Gaussian Distribution.

Generalized CLTs

Under some rather loose constraints, the average [behavior] of random variables that are not iid (i.e., not independent, and/or are not identically distributed), also follows the Gaussian Distribution.

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Central Limit Theorem

Mathematically…

Basic CLT

For iid random variables $\{X_i\}$, the average defined as

$$Z_n = \sqrt{n}\left(\frac{1}{n}\sum_{i}^{n} X_i\right)$$

Follows the Gaussian distribution in the limiting case as $n \to \infty$

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Generalized CLTs HW!

Gaussian Distribution

…the formula

A random variable X is said to be Normally Distributed if its density function follows the following formulation (called Gaussian Distribution)

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

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Shows that the distribution has two parameters.

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Where the "bell" shape comes from?

- Note that the function decays exponentially as we move away from the mean.
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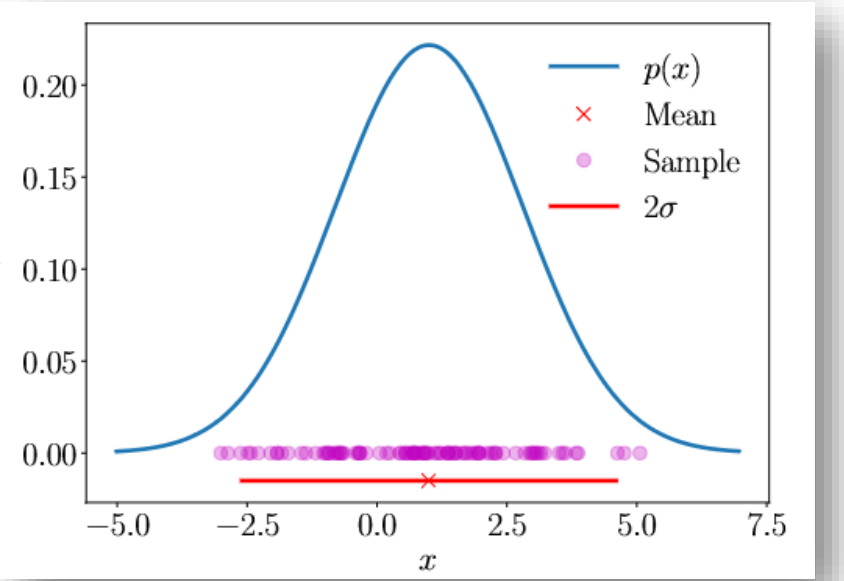
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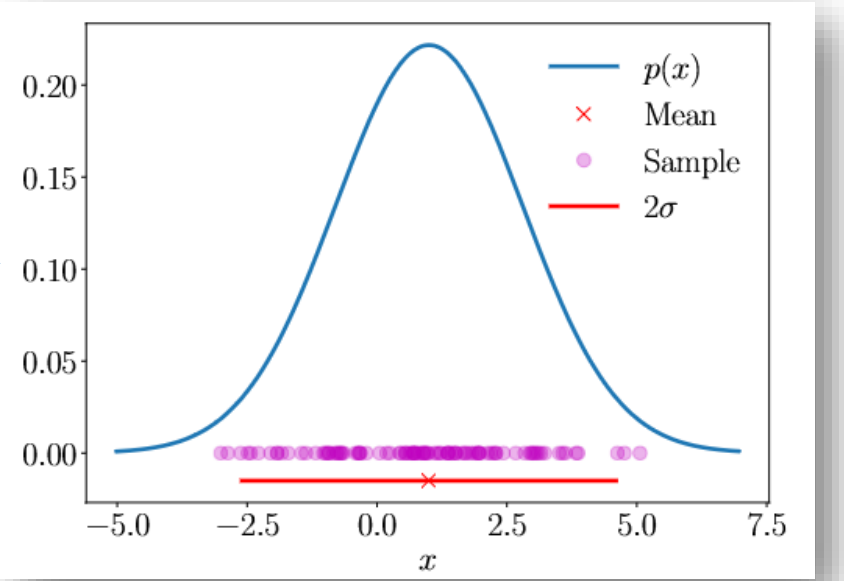
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Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$.

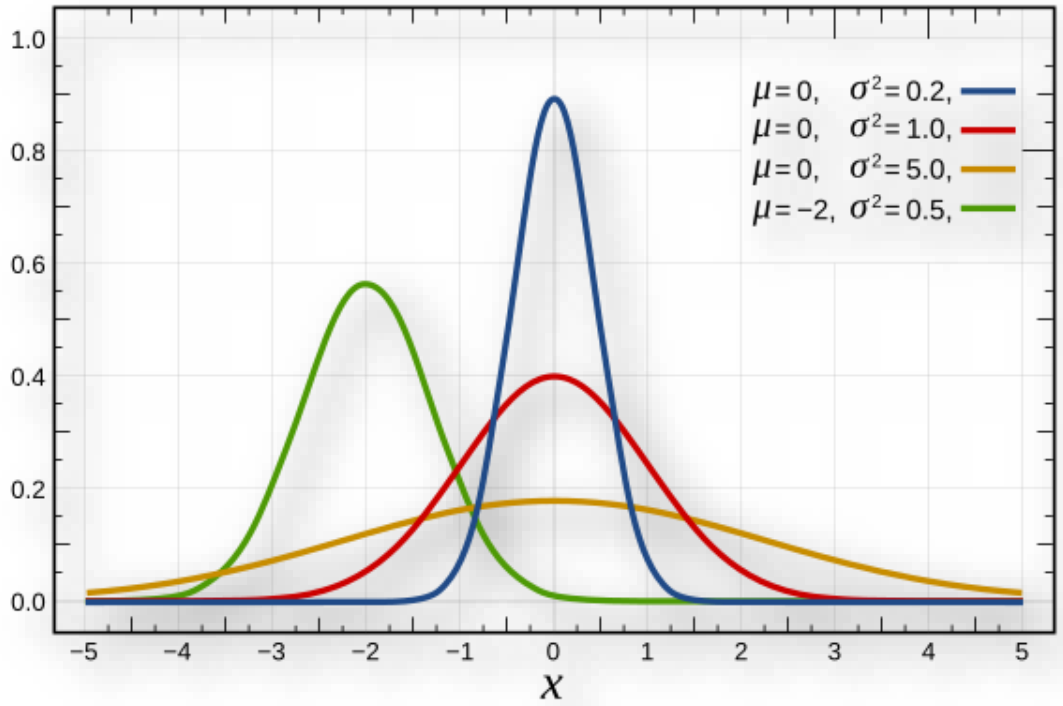
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Gaussian Distribution

Role of Mean and Variance



- "Mean" translates the distribution along the x-axis.
- "Variance" reflects the spread of the distribution.
- Q. Why does the peak go down as the variance increases?

A Gaussian Distribution is fully characterized by the mean and the variance!

Gaussian Distribution

Role of Mean and Variance

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In fact, it can be shown that all the higher moments of the Gaussian Distribution are either zero or functions of the first two moments (μ and σ^2)

$$\mathrm{E}[(X-\mu)^p] = \begin{cases} 0 & \text{if } p \text{ is odd,} \\ \sigma^p (p-1)!! & \text{if } p \text{ is even.} \end{cases}$$

$p = 0, 1, 2, 3, \ldots$

HW: read up on the "double factorial".

Gaussian Distribution

Standard Normal

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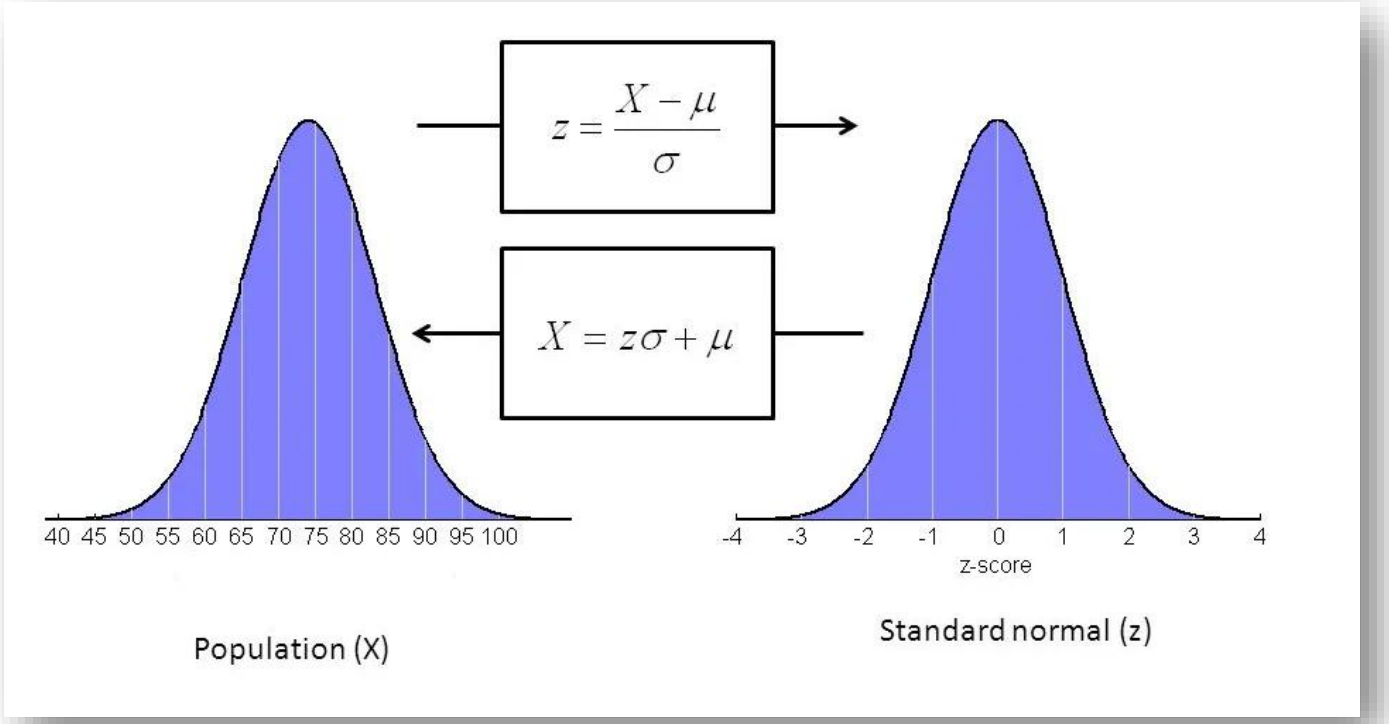
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$$Z = \frac{X - \mu}{\sigma}$$



Multivariate Gaussian Distribution

$$X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

A random vector $\boldsymbol{X} = [X_1\ X_2\ X_3\ \ldots X_D]^T$ is said to be normally distributed if it follows the multivariate Gaussian distribution given as

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Covariance matrix:
$Cov[\boldsymbol{X}] = E[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^T]$

Mean vector: $E[\boldsymbol{X}] = \left[\mu_{X_1}\ \ \mu_{X_2} \ldots\ \mu_{X_D}\right]^T$

Values whose joint likelihood we wish to calculate: $\boldsymbol{x} = [x_1\ \ x_2\ \ \ldots\ \ x_D]^T$

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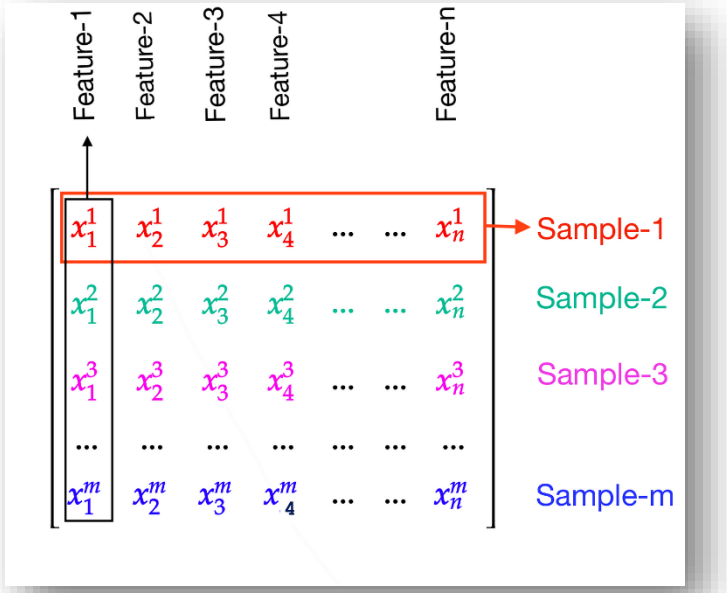
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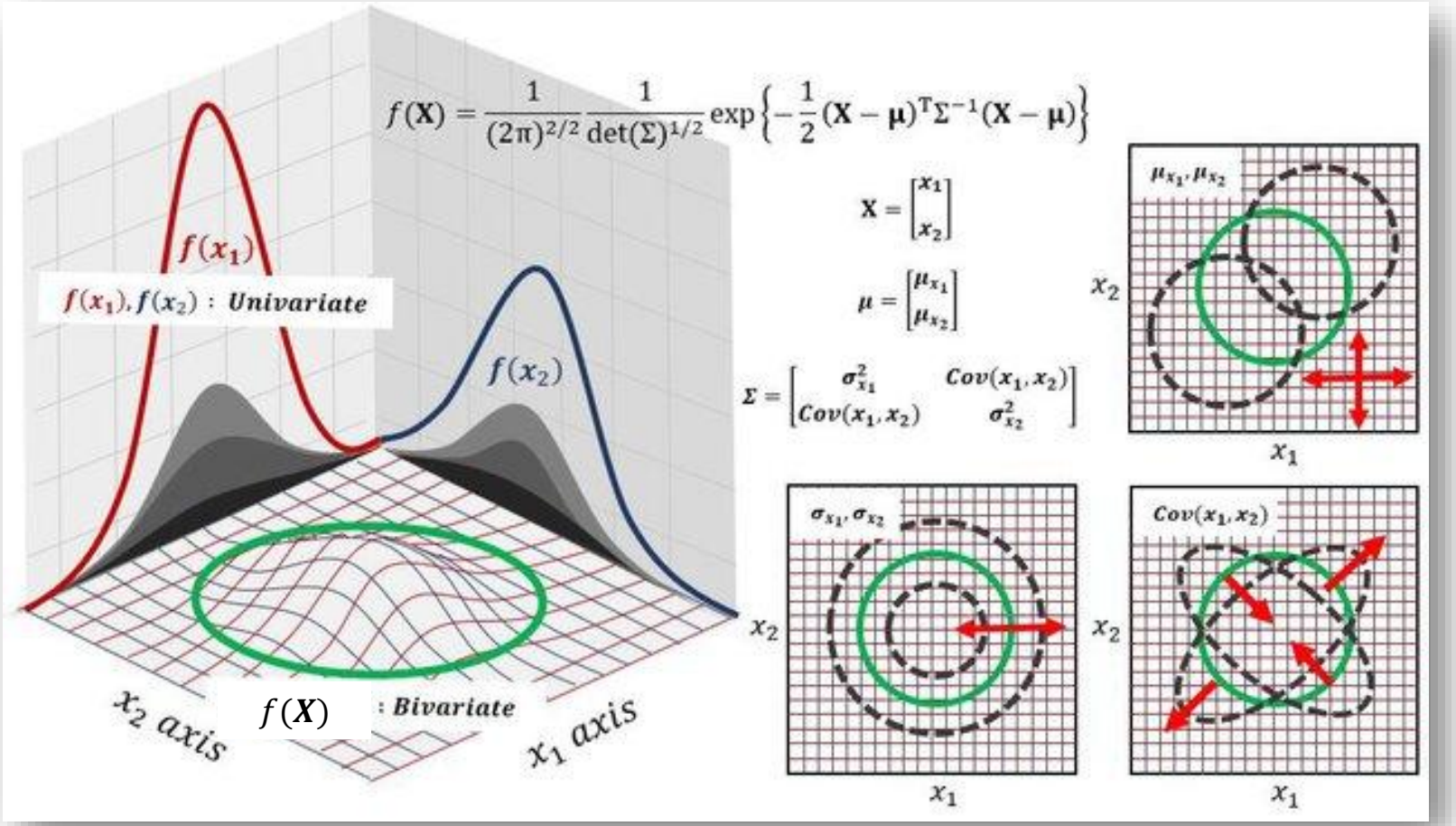
- In machine learning, we often deal with random vectors.
- Each row here is one realization of a random vector.

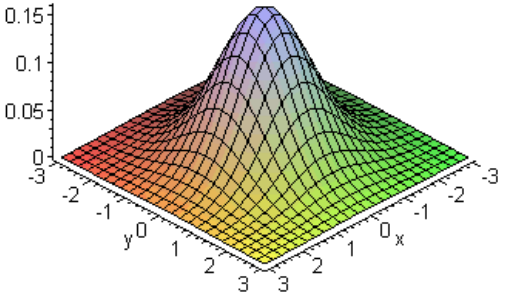


Multivariate Gaussian Distribution

Visualizing Bivariate Case…

$$\boldsymbol{X} = [X_1\ X_2\ ]^T$$

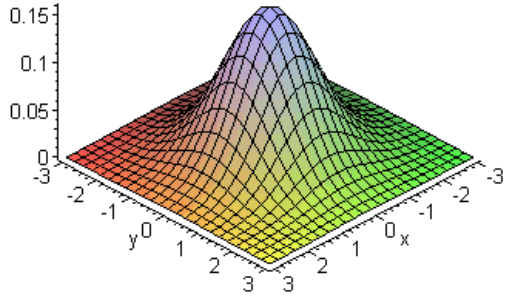




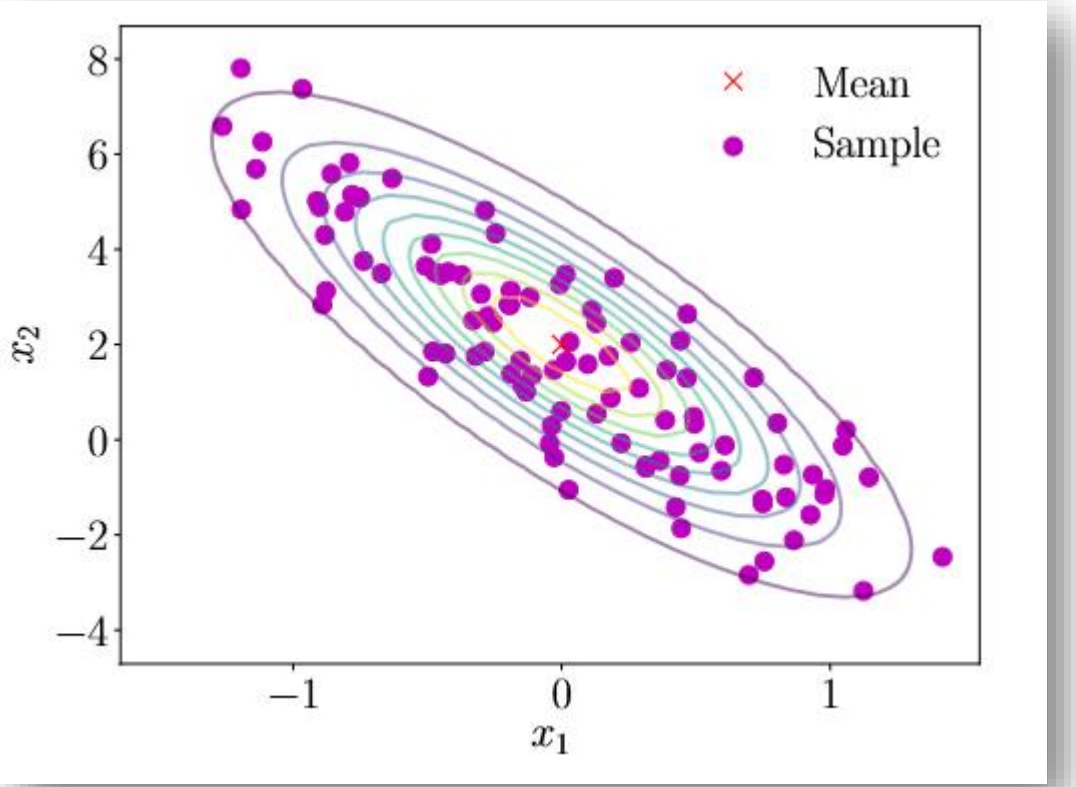
- Changing the means shifts the center.
- Changing variances varies the spread.
- Changing covariances shifts between round and elliptical with various axes.

Multivariate Gaussian Distribution

Visualizing Bivariate Case…



$$\boldsymbol{X} = [X_1\ X_2\ ]^T$$



- We often represent the distribution by its contour plots.

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Why?

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Why?

Recall the structure of the covariance matrix!

Covariance Matrix

$$\begin{bmatrix} \mathrm{Var}(x_1) & \cdots & \mathrm{Cov}(x_n, x_1) \\ \vdots & & \vdots \\ \mathrm{Cov}(x_n, x_1) & \cdots & \mathrm{Var}(x_n) \end{bmatrix}$$

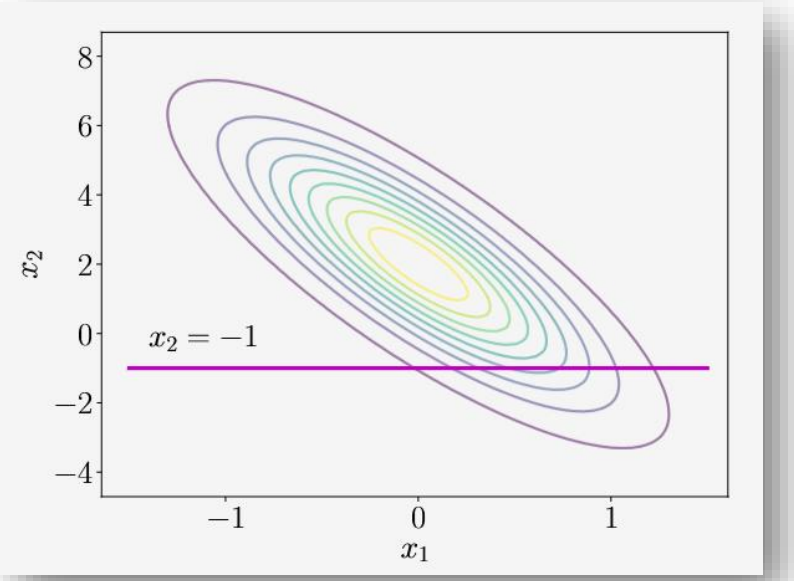
Conditional of Gaussians is also Gaussian

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Consider two random vector $\boldsymbol{x}, \boldsymbol{y}$ that are jointly Gaussian, i.e.,

$$p(\boldsymbol{x}, \boldsymbol{y}) = \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{bmatrix} \right),$$

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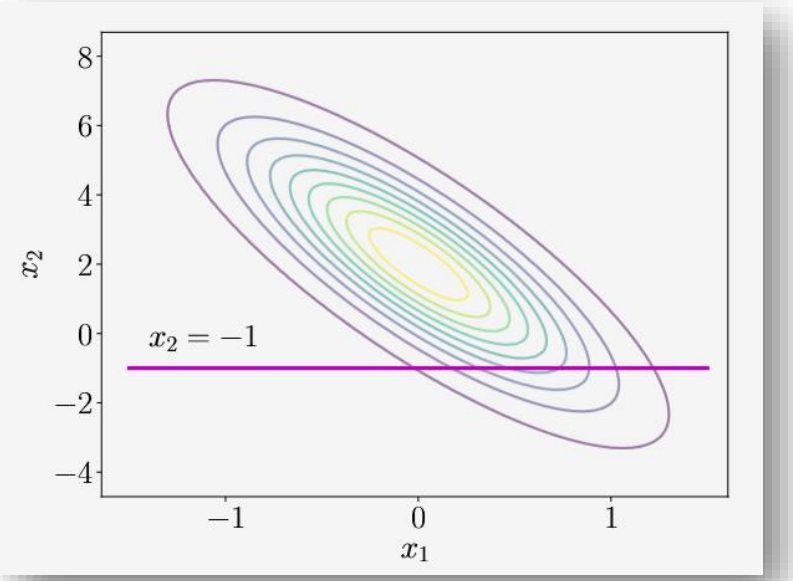


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Then, the conditional distribution $p(\boldsymbol{x}|\boldsymbol{y})$ is also Gaussian.

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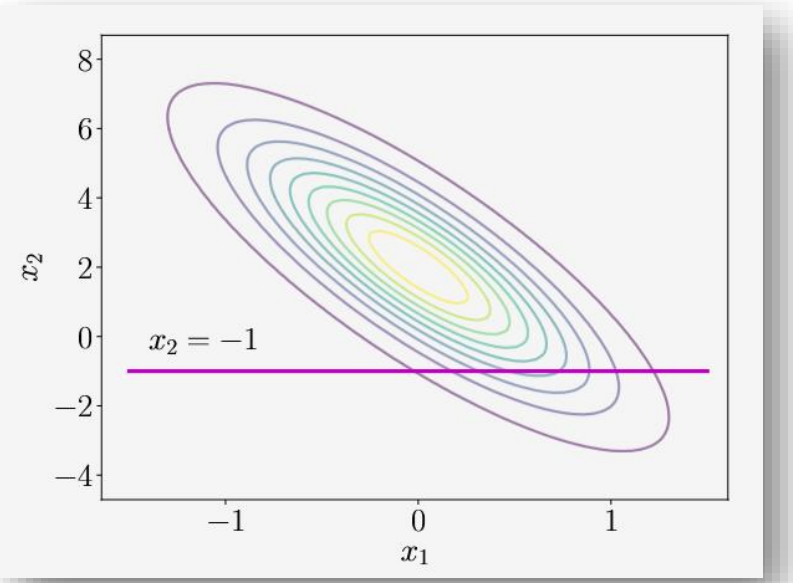
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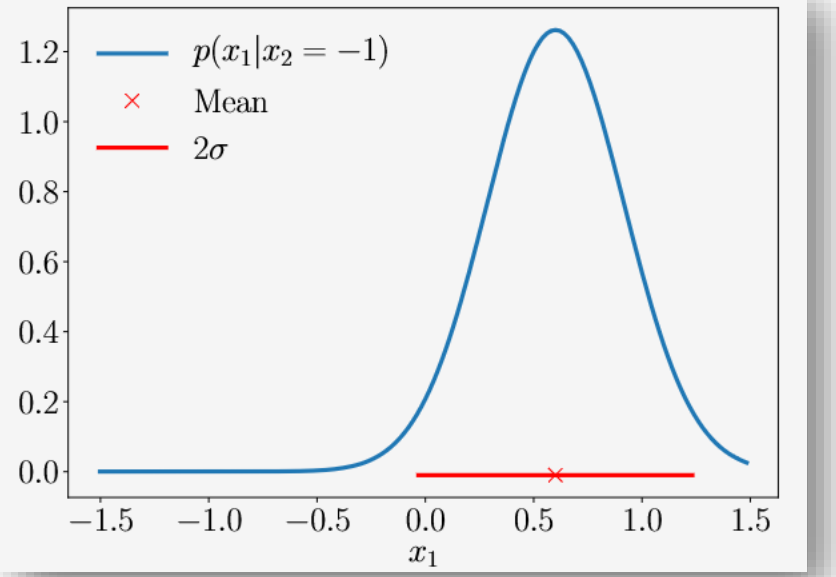
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Then, the conditional distribution $p(\boldsymbol{x}|\boldsymbol{y})$ is also Gaussian.

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Conditional of Gaussians is also Gaussian

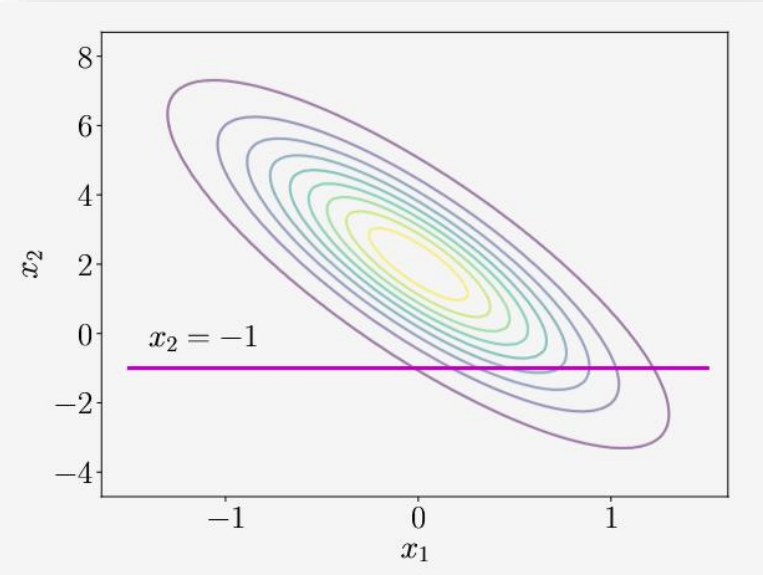
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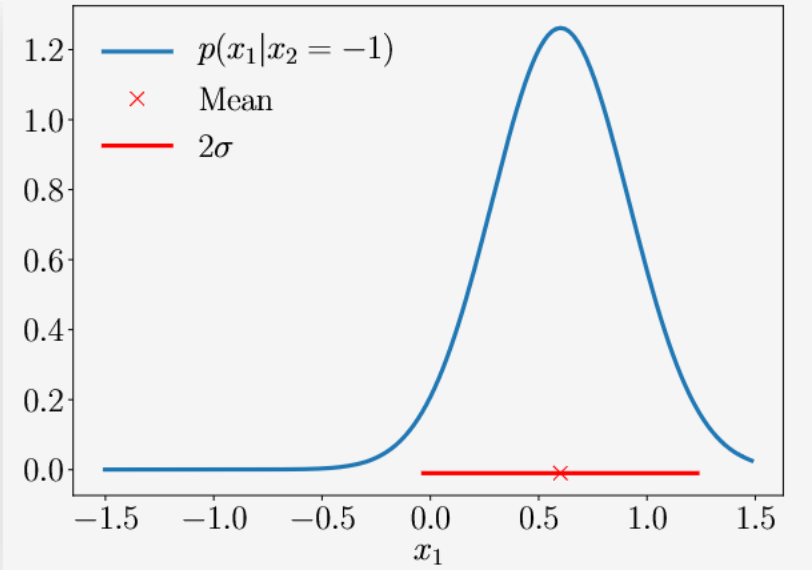
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$p(x_1|x_2 = -1)$ is the cross-section of $p(x_1, x_2)$ taken along the line $x_2 = -1$.

Marginal of Gaussians is also Gaussian

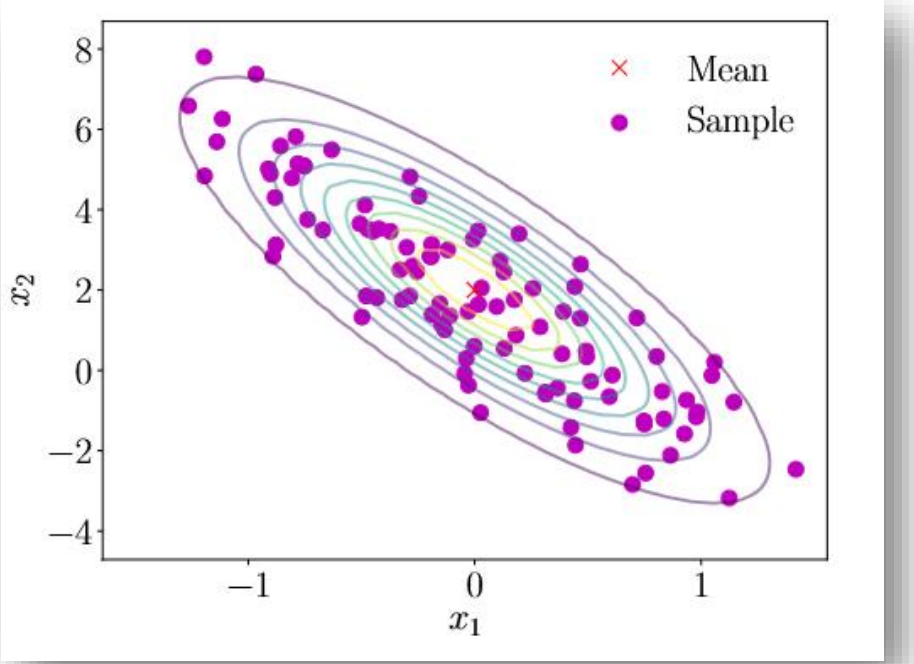
If $\boldsymbol{x}$ and $\boldsymbol{y}$ are jointly Gaussian, then the marginal distributions $p(\boldsymbol{x})$ and $p(\boldsymbol{y})$ are also Gaussian.

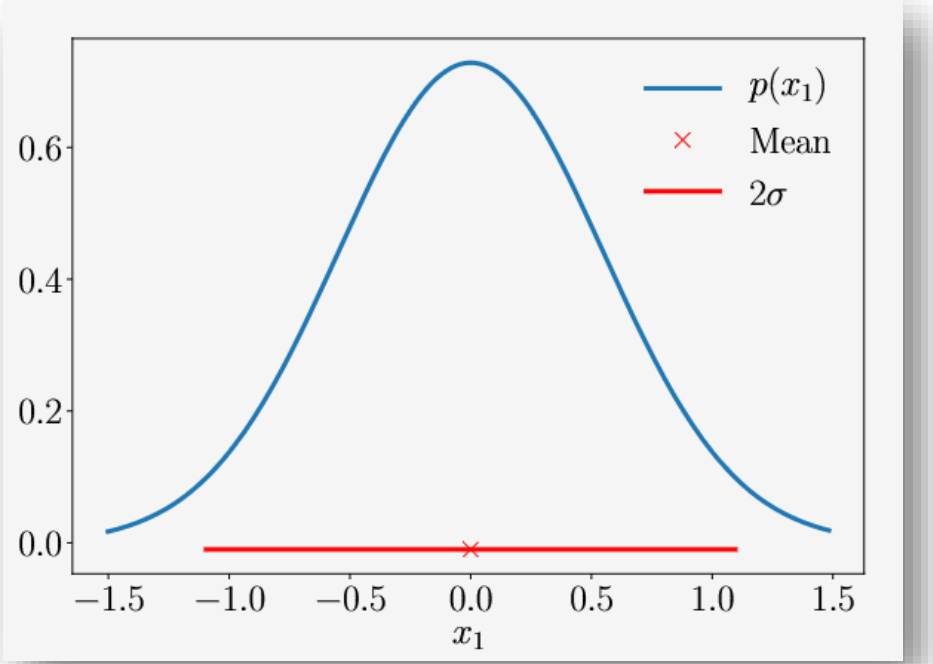
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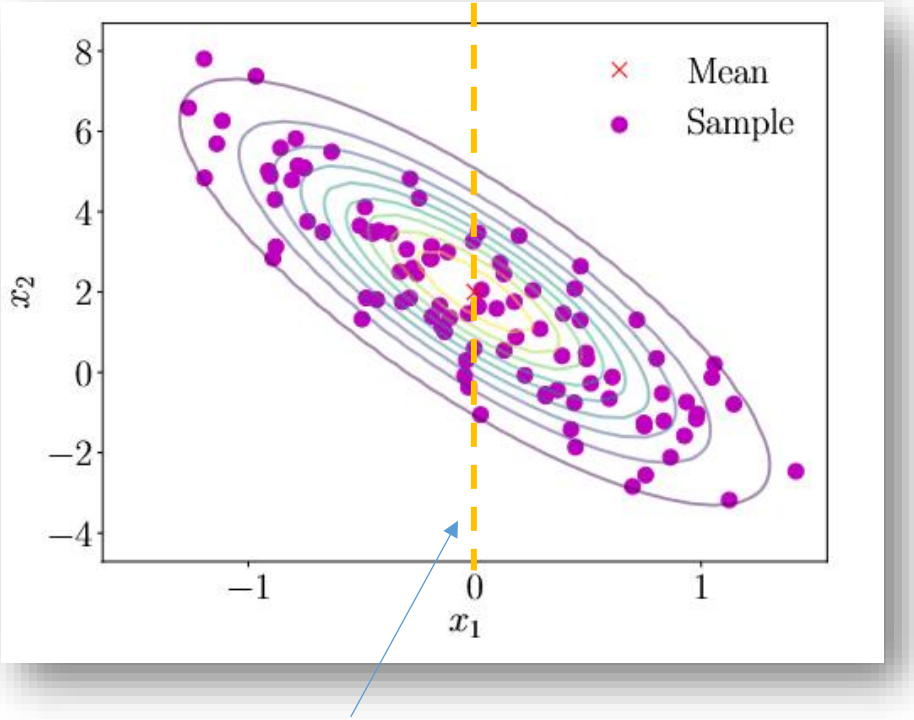


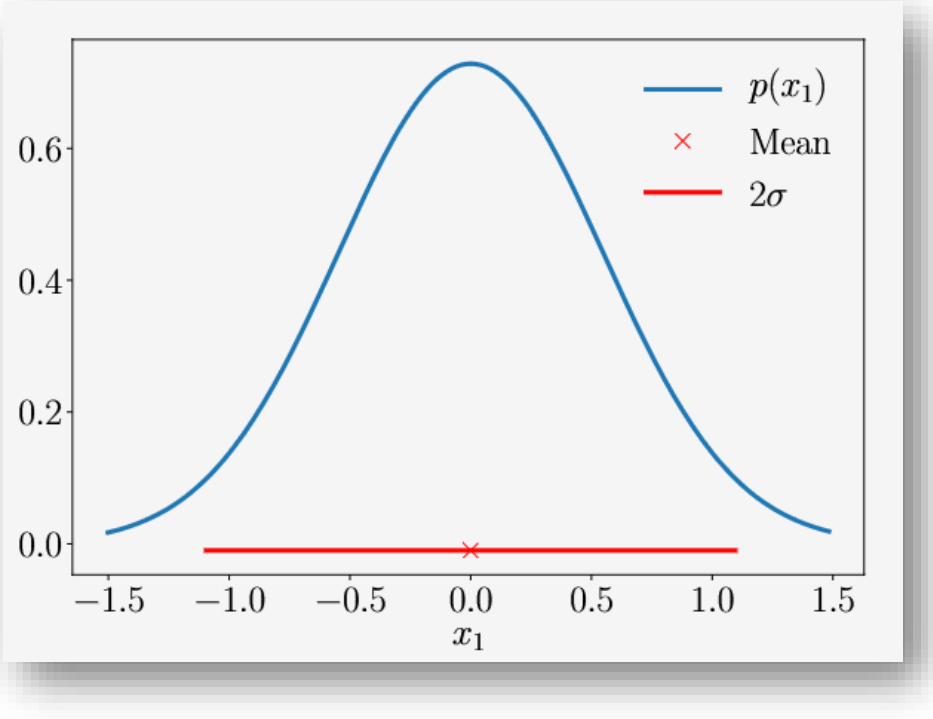


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$p(x_1 = 0)$ is the integral of $p(x_1, x_2)$ taken along this line (all possible values of x_2 against $x_1 = 0$).

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Note that the scaling factor c also follows a Gaussian-like formula (though it is not a random variable)!

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Exception 2

- The sum of two **jointly Gaussian** random variables is Gaussian (regardless of whether they are independent or not).

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$$\sigma_{X+Y} = \sqrt{\sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y},$$

$$\rho = Corr(X, Y)$$

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Finally, note that if $\boldsymbol{A}$ is invertible, the pseudoinverse reduces to inverse.

i.e., for invertible $\boldsymbol{A}$

$$(\boldsymbol{A}^T \boldsymbol{A})^{-1}\boldsymbol{A}^T = \boldsymbol{A}^{-1}\boldsymbol{A}^{-T}\boldsymbol{A}^T = \boldsymbol{A}^{-1}$$

Link Between Poisson and Gaussian Distributions

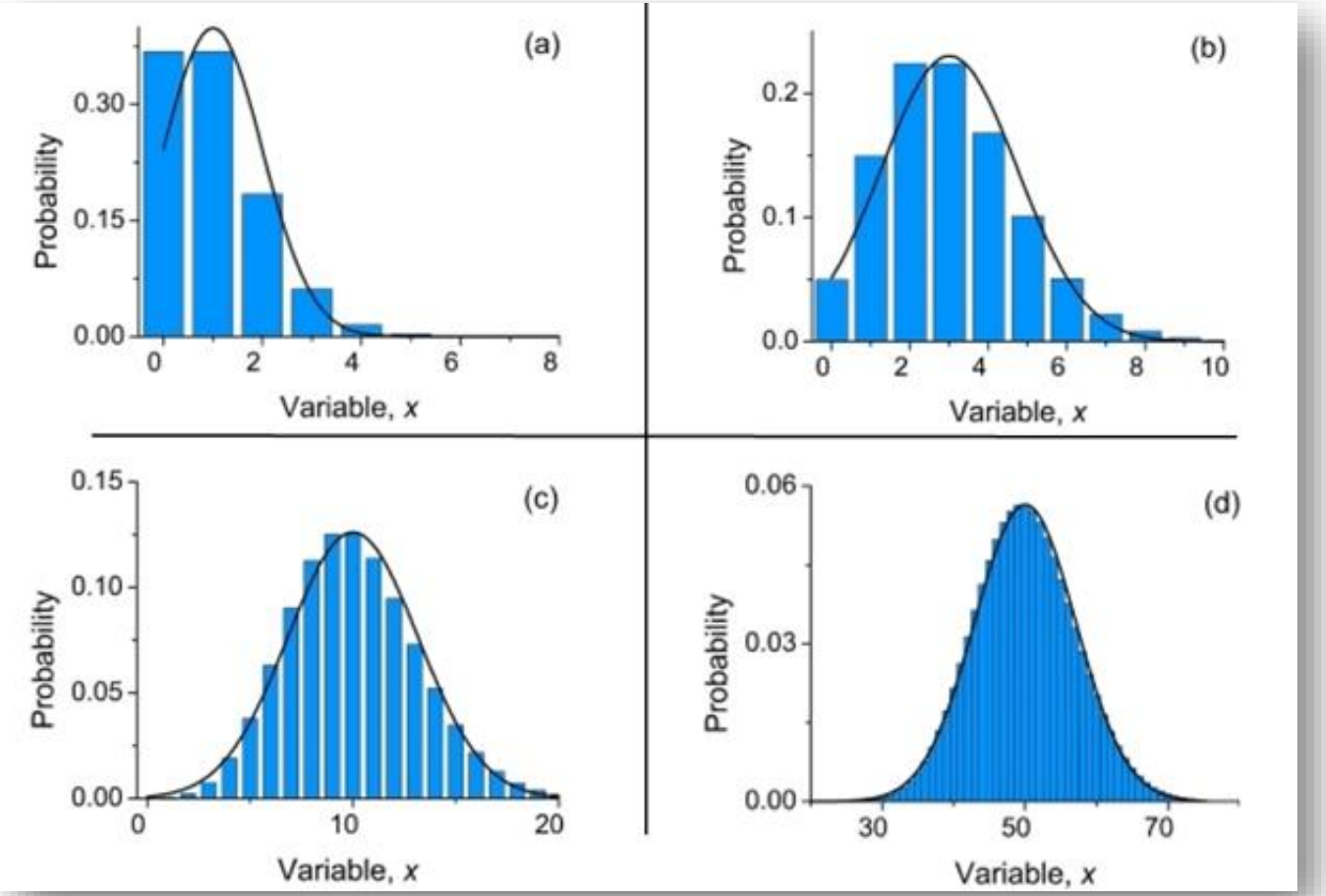
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The evolution of the Poisson distribution as λ increases. The Gaussian distribution as a function of the continuous variable x is superimposed.

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Parameter Vector

Dot Product

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Example: Showing that Gaussian Distribution Belongs to The Exponential Family.

$$f_X(x \mid \boldsymbol{\eta}) = h(x)\, \exp\Big(\boldsymbol{\eta} \cdot \mathbf{T}(x) - A(\boldsymbol{\eta})\Big),$$

$$f(y;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2}.$$

This is an exponential family which can be written in canonical form by defining

$$\boldsymbol{\eta} = \begin{bmatrix} \dfrac{\mu}{\sigma^2}, & -\dfrac{1}{2\sigma^2} \end{bmatrix}$$

$$h(y) = \frac{1}{\sqrt{2\pi}}$$

$$T(y) = \left(y, y^2\right)^{\mathrm{T}}$$

$$A(\boldsymbol{\eta}) = \frac{\mu^2}{2\sigma^2} + \log|\sigma| = -\frac{\eta_1^2}{4\eta_2} + \frac{1}{2}\log\left|\frac{1}{2\eta_2}\right|$$

In this particular case, we assume that mean and variance are unknown parameters.

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- chi-squared
- beta
- Dirichlet
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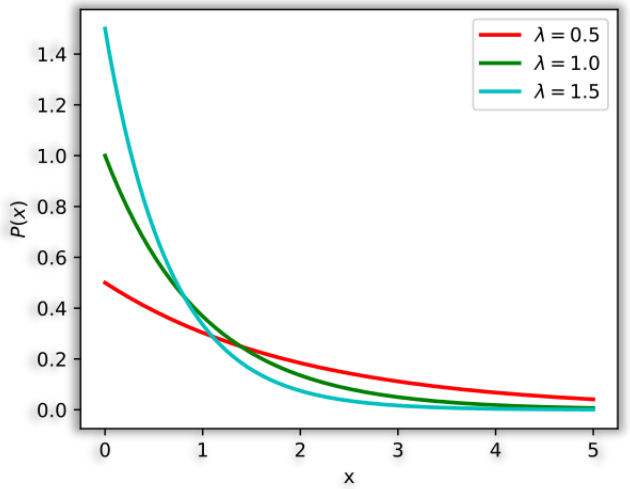
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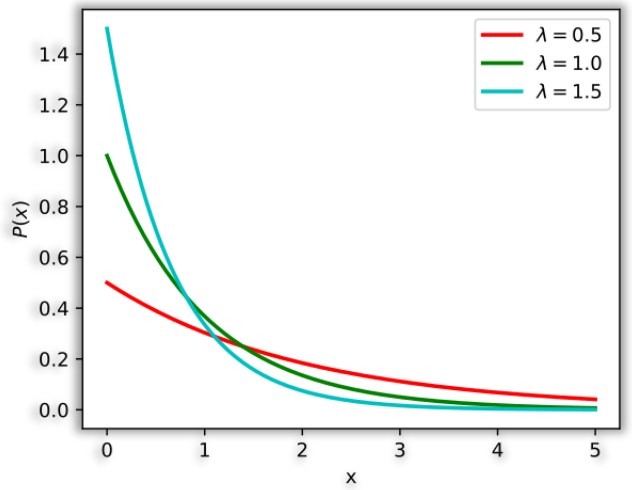
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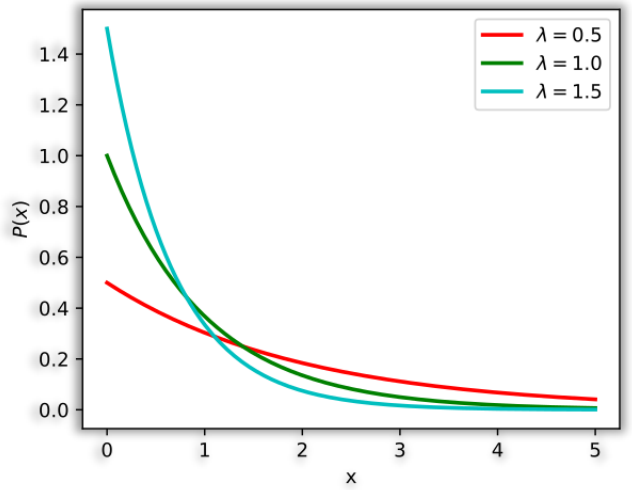
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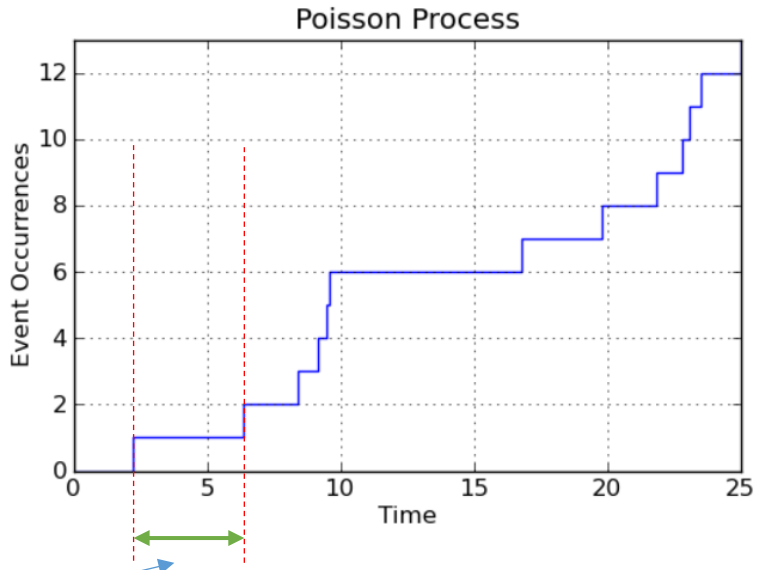
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    - E.g., the time taken for a Poisson Process to change state (count up) follows an Exponential Distribution!



Exponential Distribution

Mathematically speaking…

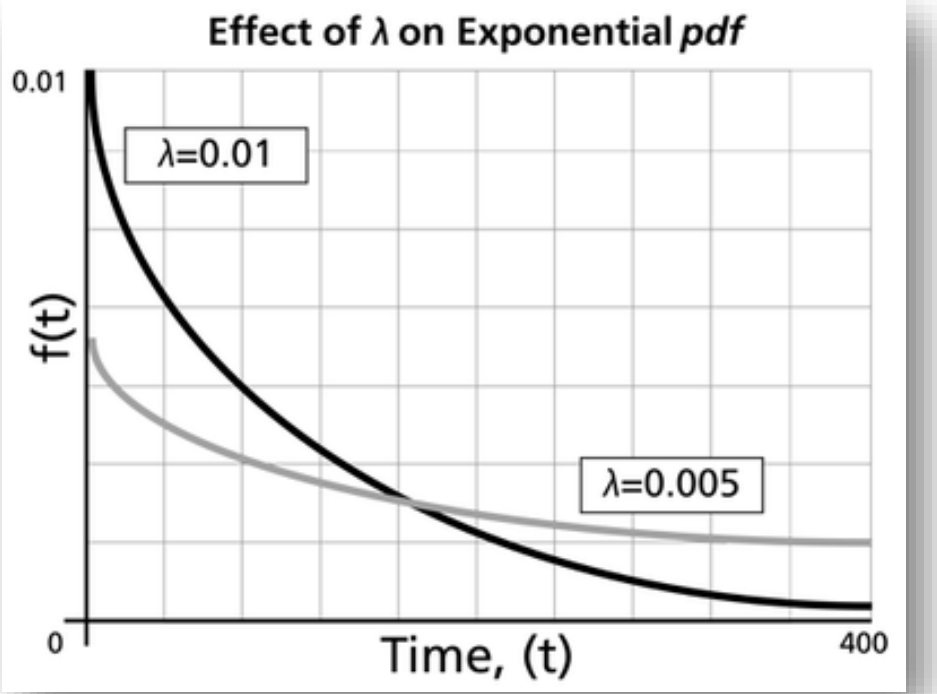
$$X \sim \text{Exp}(\lambda).$$

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases} \qquad \lambda > 0$$

$$F(x;\lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

$$\mathrm{E}[X] = \frac{1}{\lambda}.$$

$$\mathrm{Var}[X] = \frac{1}{\lambda^2},$$



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Univariate Case

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Questions?? Thoughts??

